The three-ρ terms proportional to $\log(\vee/\Lambda)$: check of the list, fixes, and the two-ρ terms

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1 Question and answer

Question. You sent all three-ρ terms proportional to $\log(\vee/\Lambda)$, with their c.c.:

- the two remainders d^3N_a, d^3N_b of the four-ρ terms;
- the genuine three-ρ terms of Group I Rows II–V, Group II Row I and Group III Rows I, II, III, V, VI.

In total there are 160 terms, written in common variables with kernel classes $\alpha, \alpha', \beta, \beta', \gamma, \gamma'$. You asked three things:

1. Do they add up to zero after reordering (symmetrizing)?
2. If not, fix them so that they do.
3. Give the corresponding two-ρ terms, which go to JIMWLK and BFKL.

Answer.

1. **As sent, the terms do not cancel** (Sec. 3).
   - Pointwise in z, all 36 kernel monomials are left over.
   - Grouped into colour networks with exact coefficients, 24 of 50 networks do not cancel.
   - The list agrees with the earlier transcription, as written, except for three typos. So the two fixes found before are still needed.

2. **Five changes make them cancel exactly** (Sec. 4). After them, all 40 colour networks cancel with exact coefficients, and the pointwise residual is exactly zero (Table 1).

   T1 Group III, Row I, term 15 (β'): $\rho^h(x') \to \rho^c(x')$. As written, the index h occurs three times. Term 11 of the same row (its α' partner) has $\rho^c(x')$.

   T2 Group III, Row II, term 13 (β'): $U^{hl}(y) \to U^{hl}(x)$. Term 9 (its α' partner) has $U^{hl}(x)$.

   T3 Group I, Row II, part 1, the eight α terms: the kernel is written $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-w)\cdot\mathcal{K}(y-z)$. The α kernel is $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)$: $\mathcal{K}(x'-w)\to\mathcal{K}(x-z)$.

   F4 Group I, Row II: remove the α terms of the $U(w)\mathbb{A}^{(3)}$ sub-terms. These are part 1, terms 1, 2, 5, 6, and their c.c., part 2, terms 1, 2, 5, 6. This part, $\mathfrak{F}_\alpha$, is the commutator $[\mathbb{N}^{(2)},\mathbb{A}^{(1)}]$. With the p^+ ordering of the virtual factor it multiplies $-\log(\vee/k^+)$, not $\log(\vee/\Lambda)$.

   F3 Group I, Row III: remove the γ,γ' terms (part 2, edge kernel ϕ, terms 9–16) and add $\mathcal{D}$. $\mathcal{D}$ has eight terms of class β,β', written in your notation in Sec. 4.4.

   T1–T3 are typos. F3 and F4 are the two fixes of the earlier three-ρ note.

3. **The two-ρ terms need the operator order of the charges, and your list does not have it** (Sec. 6). Your list gives each term in the charge order of the symmetrized check. That is enough for the three-ρ check, but not for the two-ρ terms. There are two differences:

   (a) *c.c. terms* (the primed classes): the c.c. of an operator product reverses the order of the charges. ABC and CBA have the same symmetrized part but two-ρ terms of opposite sign. Your c.c. terms keep the order of the original term.

   (b) *Rows with an anticommutator* in the notes (Group I Row V, Group II Row I, Group III Row III): your list writes $\{B,C\}A$ as $2\,BCA$. The two have the same symmetrized part, but their two-ρ terms differ by $\frac{1}{2}\{[B,C],A\}$. In Group I Row II the anticommutator is antisymmetrized by f^{abc}, and your two orderings are exactly $\{\rho,\rho\}$.

   Reordering your list literally gives wrong two-ρ terms in 11 of the 12 sources (Table 2). Sec. 6.3 gives the corrected list again, term by term, with your colour indices, in operator order; the changed terms are blue. Its two-ρ terms equal those of the original expressions exactly, row by row.

4. **The two-ρ terms** are listed source by source in Sec. 7. Added to the genuine two-ρ terms of `Evolution_two_rho.tex` and to the lists of `4rho.tex`, they give JIMWLK⊗LO exactly: 13 kernel structures, all 50 independent colour structures with a nonzero entry, exact coefficients (Sec. 8).

5. **BFKL.** BFKL⊗LO is the coefficient of $\log\vee$. To get it, the $\log(\vee/k^+)$ terms of each row are added to these $\log(\vee/\Lambda)$ terms; the result is BFKL⊗LO exactly. That part is unchanged by this check and is written out in `TwoRho_all.pdf`, Sec. 9.

2 Conventions

Units. $\mathcal{N} \equiv \frac{1}{(2\pi)^3}\frac{g^4}{16\pi^5}\frac{1}{k^+}\log\frac{\vee}{\Lambda}$, so

- your prefactor $\frac{1}{(2\pi)^3}\frac{ig^4}{16\pi^5}\frac{1}{k^+}\log\frac{\vee}{\Lambda}$ is $i\mathcal{N}$;
- $\mp\frac{i}{(2\pi)^3}\frac{g^4}{8\pi^3}\frac{\log(\vee/\Lambda)}{2\pi^2}\frac{1}{k^+}$ is $\mp i\mathcal{N}$.

Every block is (prefactor) $\int e^{-ik\cdot(w'-w)}[\dots]$ with all positions except w, w' integrated.

Kernel classes (your labels). With $\mathcal{K}(X) = X/X^2$,

$$K_\alpha = \mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z), \qquad K_{\alpha'} = \mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z),$$
$$K_\beta = \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z), \qquad K_{\beta'} = \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z),$$
$$K_\gamma = \mathcal{K}(x'-w')\cdot\mathcal{K}(y-z)\,\phi_w(x), \qquad K_{\gamma'} = \mathcal{K}(x-w)\cdot\mathcal{K}(y-z)\,\phi_{w'}(x'),$$

with $\phi_w(x)$ the edge kernel of Group I, Row III, part 2. The charge tied to w is x, the one tied to w' is x', and the third charge is y. In the lists below, term numbers are those of your list, block by block.

Symmetrization and two-ρ terms. For three charges A, B, C, with $S(ABC)$ the fully symmetrized product,

$$ABC = S(ABC) + \tfrac{1}{4}\big(\{[A,B],C\} + \{[A,C],B\} + \{[B,C],A\}\big) + (\text{one } \rho), \tag{2.1}$$

$$A\{B,C\} = 2S(ABC) + \tfrac{1}{2}\{[A,B],C\} + \tfrac{1}{2}\{[A,C],B\} + (\text{one } \rho), \tag{2.2}$$

$$\{B,C\}A = 2S(ABC) - \tfrac{1}{2}\{[A,B],C\} - \tfrac{1}{2}\{[A,C],B\} + (\text{one } \rho), \tag{2.3}$$

with $[\rho^a(x), \rho^b(y)] = i f^{abc}\rho^c(x)\delta^{(2)}(x-y)$. The three-$\rho$ check uses only S; the two-ρ terms are the anticommutators. Everything uses $U^{bc} = U^{cb}$.

3 The list as sent

Each term is symmetrized (charges commute). The sum is evaluated pointwise at random positions, random symmetric SU(3) Wilson lines and random charges. It is sorted by kernel monomial, with the charges placed on three slots in all 3! ways. Table 1 shows the result as sent and after each fix.

state	terms	kernel monomials	non-cancelling (3 configurations)	largest residual
as sent	160	36	36, 36, 36	8.3
typos T1-T3 fixed	160	36	36, 36, 36	4.6
typos + F4	152	36	24, 24, 24	4.6
typos + F3	160	24	12, 12, 12	2.6
typos + F3 + F4	152	24	0, 0, 0	0

Table 1: Pointwise test of the symmetrized three-ρ terms. "As sent" already reads the α kernel of Group I Row II part 1 as K_α (T3); with the kernel as written, those eight terms match nothing. Only the last line cancels.

Row by row. Each row of the list was compared with the earlier transcription of `Evolution_three_rho.tex` and of the four-ρ remainders (the ThreeRho_cancellation note), term by term.

- All rows agree exactly with the earlier list "as written", except Group III Rows I and II, which differ by the single terms T1 and T2.
- Row II contains the full $\mathfrak{F}$ in all four sub-terms, and Row III contains part 2 (γ, γ'). So neither F3 nor F4 is in the list.

The earlier note found that these two fixes are needed and sufficient. The colour-network tables of Sec. 5 show exactly which networks they repair.

4 The fixes

4.1 T1, T2: two index typos in Group III

T1: Group III, Row I, term 15 (β').

$$\begin{aligned}\text{as sent:}\quad & +2\,K_{\beta'}\, f^{abc}\,\rho^l(x)\,\rho^e(y)\,\rho^h(x')\,U^{de}(y)\,U^{hl}(w)\,U^{db}(z)\,U^{ha}(w')\\ \text{corrected:}\quad & +2\,K_{\beta'}\, f^{abc}\,\rho^l(x)\,\rho^e(y)\,\rho^c(x')\,U^{de}(y)\,U^{hl}(w)\,U^{db}(z)\,U^{ha}(w')\end{aligned}$$

T2: Group III, Row II, term 13 (β').

$$\begin{aligned}\text{as sent:}\quad & -2\,K_{\beta'}\, f^{abc}\,\rho^e(y)\,\rho^l(x)\,\rho^c(x')\,U^{de}(y)\,U^{hl}(y)\,U^{bd}(z)\,U^{ha}(w')\\ \text{corrected:}\quad & -2\,K_{\beta'}\, f^{abc}\,\rho^e(y)\,\rho^l(x)\,\rho^c(x')\,U^{de}(y)\,U^{hl}(x)\,U^{bd}(z)\,U^{ha}(w')\end{aligned}$$

In T1 the index h appears three times as written (in $U^{hl}(w)$, $U^{ha}(w')$, $\rho^h(x')$). The free index of the charge at x' must be c, the third index of f^{abc}, as in the α' term 11 of the same row. In T2 the Wilson line U^{hl} belongs to the charge $\rho^l(x)$, as in the α' term 9.

4.2 T3: the α kernel of Group I, Row II, part 1

All eight α terms of part 1 are written with $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-w)\cdot\mathcal{K}(y-z)$. The Fourier transform of Row II gives $\mathfrak{F}^i = \mathcal{K}^i(x-w)\int_z[K(w,y,z) - K(x,y,z)] - (x \leftrightarrow y)$. Its α part is $K(x,y,z) = \mathcal{K}(x-z)\cdot\mathcal{K}(y-z)$, and the label α says the same. So the kernel is K_α; the α' terms of part 2 (the c.c.) have the correct $K_{\alpha'}$.

4.3 F4: Group I, Row II

Split $\mathfrak{F}^i = \mathfrak{F}^i_\beta + \mathfrak{F}^i_\alpha$, with $\mathfrak{F}^i_\alpha = -\big[\mathcal{K}^i(x-w) - \mathcal{K}^i(y-w)\big] \int_z K(x,y,z)$.

- In the $U(w)\mathbb{A}^{(3)}$ sub-terms ($U^{de}(x')U^{ad}(w)$ and $U^{de}(w')U^{ad}(w)$), $\mathfrak{F}_\alpha$ is the commutator $[\mathbb{N}^{(2)}, \mathbb{A}^{(1)}]$ of the normalization with the LO amplitude (combined note, Part I).
- Ordering the virtual factor in p^+ puts its soft part to the left of $\mathbb{A}^{(1)}$ and its hard part to the right. Then this commutator comes with $-\log(\vee/k^+)$, not with $\log(k^+/\Lambda)$, so it is not a $\log(\vee/\Lambda)$ term.
- The $\mathfrak{F}_\alpha$ part of the $\bar{\mathbb{A}}^{(3)}$ sub-terms stays.

In your list this removes:

$$
\begin{aligned}
& i\mathcal{N} \int e^{-ik\cdot(w'-w)} \Big[\\
& \text{(I.II.1.1)} \quad + K_\alpha\, f^{abc}\, \rho^e(x')\, \rho^b(x)\, \rho^c(y)\, U^{de}(x')\, U^{ad}(w) \\
& \text{(I.II.1.2)} \quad - K_\alpha\, f^{abc}\, \rho^e(x')\, \rho^b(y)\, \rho^c(x)\, U^{de}(x')\, U^{ad}(w) \\
& \text{(I.II.1.5)} \quad - K_\alpha\, f^{abc}\, \rho^e(x')\, \rho^b(x)\, \rho^c(y)\, U^{de}(w')\, U^{ad}(w) \\
& \text{(I.II.1.6)} \quad + K_\alpha\, f^{abc}\, \rho^e(x')\, \rho^b(y)\, \rho^c(x)\, U^{ed}(w')\, U^{ad}(w) \\
& \text{(I.II.2.1)} \quad - K_{\alpha'}\, f^{abc}\, \rho^e(x)\, \rho^b(x')\, \rho^c(y)\, U^{de}(x)\, U^{ad}(w') \\
& \text{(I.II.2.2)} \quad + K_{\alpha'}\, f^{abc}\, \rho^e(x)\, \rho^b(y)\, \rho^c(x')\, U^{de}(x)\, U^{ad}(w') \\
& \text{(I.II.2.5)} \quad + K_{\alpha'}\, f^{abc}\, \rho^e(x)\, \rho^b(x')\, \rho^c(y)\, U^{de}(w)\, U^{ad}(w') \\
& \text{(I.II.2.6)} \quad - K_{\alpha'}\, f^{abc}\, \rho^e(x)\, \rho^b(y)\, \rho^c(x')\, U^{de}(w)\, U^{ad}(w') \Big]
\end{aligned}
$$

4.4 F3: Group I, Row III

Part 2 of Row III (the edge kernel ϕ, classes γ, γ') has nothing to cancel against. It is replaced by

$$
\begin{aligned}
\mathcal{D} = & -2i\mathcal{N} \int e^{-ik\cdot(w'-w)}\, \mathcal{K}(x'-w') \cdot \mathcal{K}(y-w)\, \mathcal{K}(x-z) \cdot \mathcal{K}(w-z) \\
& \times f^{abd} \big[U^{ab'}(x') - U^{ab'}(w')\big] \rho^{b'}(x')\, U^{de}(x) \rho^e(x) \big[U^{bc}(w) - U^{bc}(y)\big] \rho^c(y) + \text{c.c.}
\end{aligned} \tag{4.1}
$$

- $\mathcal{D}$ is Row III in the region where the gluon of $\mathbb{A}^{(1)}(p^+, z)$ carries almost all of k^+ and becomes the measured gluon. There $U(z) \to U(w)$.
- The charge y emits the measured gluon at LO ($\mathcal{K}(y-w)$), and a soft gluon at z is exchanged between the measured gluon and $U(x)\rho(x)$ ($\mathcal{K}(x-z) \cdot \mathcal{K}(w-z)$).
- In common variables ($x \leftrightarrow y$) it is of class β, and its c.c. of class β'.

In your list this removes:

$$
\begin{aligned}
& i\mathcal{N} \int e^{-ik\cdot(w'-w)} \Big[\\
& \text{(I.III.9)} \quad + 2\, K_\gamma\, f^{abc}\, \rho^d(x')\, \rho^e(x)\, \rho^h(y)\, U^{ad}(x')\, U^{ce}(x)\, U^{bh}(z) \\
& \text{(I.III.10)} \quad - 2\, K_\gamma\, f^{abc}\, \rho^d(x')\, \rho^e(x)\, \rho^h(y)\, U^{ad}(x')\, U^{ce}(x)\, U^{bh}(y) \\
& \text{(I.III.11)} \quad - 2\, K_\gamma\, f^{abc}\, \rho^d(x')\, \rho^e(x)\, \rho^h(y)\, U^{ad}(w')\, U^{ce}(x)\, U^{bh}(z) \\
& \text{(I.III.12)} \quad + 2\, K_\gamma\, f^{abc}\, \rho^d(x')\, \rho^e(x)\, \rho^h(y)\, U^{ad}(w')\, U^{ce}(x)\, U^{bh}(y) \\
& \text{(I.III.13)} \quad - 2\, K_{\gamma'}\, f^{abc}\, \rho^d(x)\, \rho^e(x')\, \rho^h(y)\, U^{ad}(x)\, U^{ce}(x')\, U^{bh}(z) \\
& \text{(I.III.14)} \quad + 2\, K_{\gamma'}\, f^{abc}\, \rho^d(x)\, \rho^e(x')\, \rho^h(y)\, U^{ad}(x)\, U^{ce}(x')\, U^{bh}(y) \\
& \text{(I.III.15)} \quad + 2\, K_{\gamma'}\, f^{abc}\, \rho^d(x)\, \rho^e(x')\, \rho^h(y)\, U^{ad}(w)\, U^{ce}(x')\, U^{bh}(z) \\
& \text{(I.III.16)} \quad - 2\, K_{\gamma'}\, f^{abc}\, \rho^d(x)\, \rho^e(x')\, \rho^h(y)\, U^{ad}(w)\, U^{ce}(x')\, U^{bh}(y) \Big]
\end{aligned}
$$

and adds, in your notation, the eight terms of $\mathcal{D}$:

$$
\begin{aligned}
& i\mathcal{N} \int e^{-ik\cdot(w'-w)} \Big[\\
& (1) \quad - 2\, K_\beta\, f^{abc}\, \rho^d(x')\, \rho^e(y)\, \rho^h(x)\, U^{ad}(x')\, U^{ce}(y)\, U^{bh}(w) \\
& (2) \quad + 2\, K_\beta\, f^{abc}\, \rho^d(x')\, \rho^e(y)\, \rho^h(x)\, U^{ad}(x')\, U^{ce}(y)\, U^{bh}(x) \\
& (3) \quad + 2\, K_\beta\, f^{abc}\, \rho^d(x')\, \rho^e(y)\, \rho^h(x)\, U^{ad}(w')\, U^{ce}(y)\, U^{bh}(w) \\
& (4) \quad - 2\, K_\beta\, f^{abc}\, \rho^d(x')\, \rho^e(y)\, \rho^h(x)\, U^{ad}(w')\, U^{ce}(y)\, U^{bh}(x)
\end{aligned}
$$

$$
\begin{aligned}
(5)\quad & +2\,K_{\beta'}\,f^{abc}\,\rho^d(x)\,\rho^e(y)\,\rho^h(x')\,U^{ad}(x)\,U^{ce}(y)\,U^{bh}(w')\\
(6)\quad & -2\,K_{\beta'}\,f^{abc}\,\rho^d(x)\,\rho^e(y)\,\rho^h(x')\,U^{ad}(x)\,U^{ce}(y)\,U^{bh}(x')\\
(7)\quad & -2\,K_{\beta'}\,f^{abc}\,\rho^d(x)\,\rho^e(y)\,\rho^h(x')\,U^{ad}(w)\,U^{ce}(y)\,U^{bh}(w')\\
(8)\quad & +2\,K_{\beta'}\,f^{abc}\,\rho^d(x)\,\rho^e(y)\,\rho^h(x')\,U^{ad}(w)\,U^{ce}(y)\,U^{bh}(x')\Big]
\end{aligned}
$$

For the two-ρ terms of $\mathcal{D}$ the operator order is that of Row III, $\rho(x')\rho(x)\rho(y)$ before the renaming $x \leftrightarrow y$. In common variables it is $\rho(x')\rho(y)\rho(x)$, and the c.c. is reversed.

4.5 The corrected list

The full list after the fixes, in your notation and in your (symmetrized) charge order.

- Removed terms are red: [F3], [F4].
- Corrected and added terms are blue: [T1], [T2], [T3], [F3].
- The kernel of a T3 term is K_α as printed.

From four ρ's, d^3N_a.

$$
\begin{aligned}
&-i\mathcal{N}\int e^{-ik\cdot(w'-w)}\Big[\\
&(1)\quad +K_\alpha\,f^{abc}\,\rho^d(x')\,\rho^a(y)\,\rho^c(x)\,U^{de}(w')\,U^{eb}(w)\\
&(2)\quad -K_\alpha\,f^{abc}\,\rho^e(x')\,\rho^a(y)\,\rho^c(x)\,U^{db}(w)\,U^{ed}(x')\\
&(3)\quad +K_{\alpha'}\,f^{abc}\,\rho^c(x')\,\rho^a(y)\,\rho^e(x)\,U^{db}(w')\,U^{ed}(x)\\
&(4)\quad -K_{\alpha'}\,f^{abc}\,\rho^c(x')\,\rho^a(y)\,\rho^e(x)\,U^{bd}(w')\,U^{ed}(w)\Big]
\end{aligned}
$$

From four ρ's, d^3N_b.

$$
\begin{aligned}
&i\mathcal{N}\int e^{-ik\cdot(w'-w)}\Big[\\
&(1)\quad -K_\alpha\,f^{abc}\,\rho^d(x')\,\rho^h(y)\,\rho^e(x)\,U^{db}(w')\,U^{ec}(x)\,U^{ah}(y)\\
&(2)\quad +K_{\alpha'}\,f^{abc}\,\rho^d(x')\,\rho^e(x)\,\rho^h(y)\,U^{dc}(x')\,U^{eb}(w)\,U^{ah}(y)\\
&(3)\quad -K_{\alpha'}\,f^{abc}\,\rho^d(x')\,\rho^e(x)\,\rho^h(y)\,U^{dc}(x')\,U^{be}(x)\,U^{ah}(y)\\
&(4)\quad +K_\alpha\,f^{abc}\,\rho^e(x')\,\rho^h(y)\,\rho^d(x)\,U^{dc}(x)\,U^{be}(x')\,U^{ah}(y)\Big]
\end{aligned}
$$

Group I, Row II, part 1.

$$
\begin{aligned}
&i\mathcal{N}\int e^{-ik\cdot(w'-w)}\Big[\\
&(1)\quad +K_\alpha\,f^{abc}\,\rho^e(x')\,\rho^b(x)\,\rho^c(y)\,U^{de}(x')\,U^{ad}(w)\ \text{[F4: removed]}\\
&(2)\quad -K_\alpha\,f^{abc}\,\rho^e(x')\,\rho^b(y)\,\rho^c(x)\,U^{de}(x')\,U^{ad}(w)\ \text{[F4: removed]}\\
&(3)\quad -K_\alpha\,f^{abc}\,\rho^d(x')\,\rho^e(x)\,\rho^h(y)\,U^{ad}(x')\,U^{be}(x)\,U^{ch}(y)\ \text{[T3]}\\
&(4)\quad +K_\alpha\,f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(x')\,U^{be}(y)\,U^{ch}(x)\ \text{[T3]}\\
&(5)\quad -K_\alpha\,f^{abc}\,\rho^e(x')\,\rho^b(x)\,\rho^c(y)\,U^{de}(w')\,U^{ad}(w)\ \text{[F4: removed]}\\
&(6)\quad +K_\alpha\,f^{abc}\,\rho^e(x')\,\rho^b(y)\,\rho^c(x)\,U^{ed}(w')\,U^{ad}(w)\ \text{[F4: removed]}\\
&(7)\quad +K_\alpha\,f^{abc}\,\rho^d(x')\,\rho^e(x)\,\rho^h(y)\,U^{ad}(w')\,U^{be}(x)\,U^{ch}(y)\ \text{[T3]}\\
&(8)\quad -K_\alpha\,f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(w')\,U^{be}(y)\,U^{ch}(x)\ \text{[T3]}\\
&(9)\quad -K_\beta\,f^{abc}\,\rho^e(x')\,\rho^b(x)\,\rho^c(y)\,U^{de}(x')\,U^{ad}(w)\\
&(10)\quad +K_\beta\,f^{abc}\,\rho^e(x')\,\rho^b(y)\,\rho^c(x)\,U^{de}(x')\,U^{ad}(w)\\
&(11)\quad +K_\beta\,f^{abc}\,\rho^d(x')\,\rho^e(x)\,\rho^h(y)\,U^{ad}(x')\,U^{be}(x)\,U^{ch}(y)\\
&(12)\quad -K_\beta\,f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(x')\,U^{be}(y)\,U^{ch}(x)\\
&(13)\quad +K_\beta\,f^{abc}\,\rho^e(x')\,\rho^b(x)\,\rho^c(y)\,U^{de}(w')\,U^{ad}(w)\\
&(14)\quad -K_\beta\,f^{abc}\,\rho^e(x')\,\rho^b(y)\,\rho^c(x)\,U^{de}(w')\,U^{ad}(w)\\
&(15)\quad -K_\beta\,f^{abc}\,\rho^d(x')\,\rho^e(x)\,\rho^h(y)\,U^{ad}(w')\,U^{be}(x)\,U^{ch}(y)\\
&(16)\quad +K_\beta\,f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(w')\,U^{be}(y)\,U^{ch}(x)\Big]
\end{aligned}
$$

Group I, Row II, part 2 (c.c.).

$$
\begin{aligned}
& i\mathcal{N} \int e^{-ik\cdot(w'-w)} \Big[\\
& (1) \quad -K_{\alpha'}\, f^{abc}\, \rho^e(x)\, \rho^b(x')\, \rho^c(y)\, U^{de}(x)\, U^{ad}(w') \text{ [F4: removed]} \\
& (2) \quad +K_{\alpha'}\, f^{abc}\, \rho^e(x)\, \rho^b(y)\, \rho^c(x')\, U^{de}(x)\, U^{ad}(w') \text{ [F4: removed]} \\
& (3) \quad +K_{\alpha'}\, f^{abc}\, \rho^d(x)\, \rho^e(x')\, \rho^h(y)\, U^{ad}(x)\, U^{be}(x')\, U^{ch}(y) \\
& (4) \quad -K_{\alpha'}\, f^{abc}\, \rho^d(x)\, \rho^e(y)\, \rho^h(x')\, U^{ad}(x)\, U^{be}(y)\, U^{ch}(x') \\
& (5) \quad +K_{\alpha'}\, f^{abc}\, \rho^e(x)\, \rho^b(x')\, \rho^c(y)\, U^{de}(w)\, U^{ad}(w') \text{ [F4: removed]} \\
& (6) \quad -K_{\alpha'}\, f^{abc}\, \rho^e(x)\, \rho^b(y)\, \rho^c(x')\, U^{de}(w)\, U^{ad}(w') \text{ [F4: removed]} \\
& (7) \quad -K_{\alpha'}\, f^{abc}\, \rho^d(x)\, \rho^e(x')\, \rho^h(y)\, U^{ad}(w)\, U^{be}(x')\, U^{ch}(y) \\
& (8) \quad +K_{\alpha'}\, f^{abc}\, \rho^d(x)\, \rho^e(y)\, \rho^h(x')\, U^{ad}(w)\, U^{be}(y)\, U^{ch}(x') \\
& (9) \quad +K_{\beta'}\, f^{abc}\, \rho^e(x)\, \rho^b(x')\, \rho^c(y)\, U^{de}(x)\, U^{ad}(w') \\
& (10) \quad -K_{\beta'}\, f^{abc}\, \rho^e(x)\, \rho^b(y)\, \rho^c(x')\, U^{de}(x)\, U^{ad}(w') \\
& (11) \quad -K_{\beta'}\, f^{abc}\, \rho^d(x)\, \rho^e(x')\, \rho^h(y)\, U^{ad}(x)\, U^{be}(x')\, U^{ch}(y) \\
& (12) \quad +K_{\beta'}\, f^{abc}\, \rho^d(x)\, \rho^e(y)\, \rho^h(x')\, U^{ad}(x)\, U^{be}(y)\, U^{ch}(x') \\
& (13) \quad -K_{\beta'}\, f^{abc}\, \rho^e(x)\, \rho^b(x')\, \rho^c(y)\, U^{de}(w)\, U^{ad}(w') \\
& (14) \quad +K_{\beta'}\, f^{abc}\, \rho^e(x)\, \rho^b(y)\, \rho^c(x')\, U^{de}(w)\, U^{ad}(w') \\
& (15) \quad +K_{\beta'}\, f^{abc}\, \rho^d(x)\, \rho^e(x')\, \rho^h(y)\, U^{ad}(w)\, U^{be}(x')\, U^{ch}(y) \\
& (16) \quad -K_{\beta'}\, f^{abc}\, \rho^d(x)\, \rho^e(y)\, \rho^h(x')\, U^{ad}(w)\, U^{be}(y)\, U^{ch}(x') \Big]
\end{aligned}
$$

Group I, Row III.

$$
\begin{aligned}
& i\mathcal{N} \int e^{-ik\cdot(w'-w)} \Big[\\
& (1) \quad +K_{\alpha}\, f^{abc}\, \rho^d(x')\, \rho^e(x)\, \rho^h(y)\, U^{ad}(x')\, U^{ce}(x)\, U^{bh}(z) \\
& (2) \quad -K_{\alpha}\, f^{abc}\, \rho^d(x')\, \rho^e(x)\, \rho^h(y)\, U^{ad}(x')\, U^{ce}(x)\, U^{bh}(y) \\
& (3) \quad -K_{\alpha}\, f^{abc}\, \rho^d(x')\, \rho^e(x)\, \rho^h(y)\, U^{ad}(w')\, U^{ce}(x)\, U^{bh}(z) \\
& (4) \quad +K_{\alpha}\, f^{abc}\, \rho^d(x')\, \rho^e(x)\, \rho^h(y)\, U^{ad}(w')\, U^{ce}(x)\, U^{bh}(y) \\
& (5) \quad -K_{\alpha'}\, f^{abc}\, \rho^d(x)\, \rho^e(x')\, \rho^h(y)\, U^{ad}(x)\, U^{ce}(x')\, U^{bh}(z) \\
& (6) \quad +K_{\alpha'}\, f^{abc}\, \rho^d(x)\, \rho^e(x')\, \rho^h(y)\, U^{ad}(x)\, U^{ce}(x')\, U^{bh}(y) \\
& (7) \quad +K_{\alpha'}\, f^{abc}\, \rho^d(x)\, \rho^e(x')\, \rho^h(y)\, U^{ad}(w)\, U^{ce}(x')\, U^{bh}(z) \\
& (8) \quad -K_{\alpha'}\, f^{abc}\, \rho^d(x)\, \rho^e(x')\, \rho^h(y)\, U^{ad}(w)\, U^{ce}(x')\, U^{bh}(y) \\
& (9) \quad +2\,K_{\gamma}\, f^{abc}\, \rho^d(x')\, \rho^e(x)\, \rho^h(y)\, U^{ad}(x')\, U^{ce}(x)\, U^{bh}(z) \text{ [F3: removed]} \\
& (10) \quad -2\,K_{\gamma}\, f^{abc}\, \rho^d(x')\, \rho^e(x)\, \rho^h(y)\, U^{ad}(x')\, U^{ce}(x)\, U^{bh}(y) \text{ [F3: removed]} \\
& (11) \quad -2\,K_{\gamma}\, f^{abc}\, \rho^d(x')\, \rho^e(x)\, \rho^h(y)\, U^{ad}(w')\, U^{ce}(x)\, U^{bh}(z) \text{ [F3: removed]} \\
& (12) \quad +2\,K_{\gamma}\, f^{abc}\, \rho^d(x')\, \rho^e(x)\, \rho^h(y)\, U^{ad}(w')\, U^{ce}(x)\, U^{bh}(y) \text{ [F3: removed]} \\
& (13) \quad -2\,K_{\gamma'}\, f^{abc}\, \rho^d(x)\, \rho^e(x')\, \rho^h(y)\, U^{ad}(x)\, U^{ce}(x')\, U^{bh}(z) \text{ [F3: removed]} \\
& (14) \quad +2\,K_{\gamma'}\, f^{abc}\, \rho^d(x)\, \rho^e(x')\, \rho^h(y)\, U^{ad}(x)\, U^{ce}(x')\, U^{bh}(y) \text{ [F3: removed]} \\
& (15) \quad +2\,K_{\gamma'}\, f^{abc}\, \rho^d(x)\, \rho^e(x')\, \rho^h(y)\, U^{ad}(w)\, U^{ce}(x')\, U^{bh}(z) \text{ [F3: removed]} \\
& (16) \quad -2\,K_{\gamma'}\, f^{abc}\, \rho^d(x)\, \rho^e(x')\, \rho^h(y)\, U^{ad}(w)\, U^{ce}(x')\, U^{bh}(y) \text{ [F3: removed]} \Big]
\end{aligned}
$$

$\mathcal{D}$ (replaces the γ, γ' terms of Group I, Row III).

$$
\begin{aligned}
& i\mathcal{N} \int e^{-ik\cdot(w'-w)} \Big[\\
& (1) \quad -2\,K_{\beta}\, f^{abc}\, \rho^d(x')\, \rho^e(y)\, \rho^h(x)\, U^{ad}(x')\, U^{ce}(y)\, U^{bh}(w) \text{ [F3]} \\
& (2) \quad +2\,K_{\beta}\, f^{abc}\, \rho^d(x')\, \rho^e(y)\, \rho^h(x)\, U^{ad}(x')\, U^{ce}(y)\, U^{bh}(x) \text{ [F3]} \\
& (3) \quad +2\,K_{\beta}\, f^{abc}\, \rho^d(x')\, \rho^e(y)\, \rho^h(x)\, U^{ad}(w')\, U^{ce}(y)\, U^{bh}(w) \text{ [F3]} \\
& (4) \quad -2\,K_{\beta}\, f^{abc}\, \rho^d(x')\, \rho^e(y)\, \rho^h(x)\, U^{ad}(w')\, U^{ce}(y)\, U^{bh}(x) \text{ [F3]} \\
& (5) \quad +2\,K_{\beta'}\, f^{abc}\, \rho^d(x)\, \rho^e(y)\, \rho^h(x')\, U^{ad}(x)\, U^{ce}(y)\, U^{bh}(w') \text{ [F3]} \\
& (6) \quad -2\,K_{\beta'}\, f^{abc}\, \rho^d(x)\, \rho^e(y)\, \rho^h(x')\, U^{ad}(x)\, U^{ce}(y)\, U^{bh}(x') \text{ [F3]} \\
& (7) \quad -2\,K_{\beta'}\, f^{abc}\, \rho^d(x)\, \rho^e(y)\, \rho^h(x')\, U^{ad}(w)\, U^{ce}(y)\, U^{bh}(w') \text{ [F3]}
\end{aligned}
$$

$$(8)\quad +2\,K_{\beta'}\,f^{abc}\,\rho^d(x)\,\rho^e(y)\,\rho^h(x')\,U^{ad}(w)\,U^{ce}(y)\,U^{bh}(x')\ \text{[F3]}\Big]$$

Group I, Row IV.

$$i\mathcal{N}\int e^{-ik\cdot(w'-w)}\Big[$$
$$\begin{aligned}
(1)\quad &+K_\alpha\,f^{abc}\,\rho^e(x')\,\rho^l(y)\,\rho^c(x)\,U^{de}(x')\,U^{ad}(w)\,U^{hb}(z)\,U^{hl}(y)\\
(2)\quad &-K_\alpha\,f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{da}(x')\,U^{be}(y)\,U^{ch}(x)\\
(3)\quad &-K_\alpha\,f^{abc}\,\rho^e(x')\,\rho^l(y)\,\rho^c(x)\,U^{de}(w')\,U^{ad}(w)\,U^{hb}(z)\,U^{hl}(y)\\
(4)\quad &+K_\alpha\,f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{da}(w')\,U^{be}(y)\,U^{ch}(x)\\
(5)\quad &-K_{\alpha'}\,f^{abc}\,\rho^e(x)\,\rho^l(y)\,\rho^c(x')\,U^{de}(x)\,U^{ad}(w')\,U^{hb}(z)\,U^{hl}(y)\\
(6)\quad &+K_{\alpha'}\,f^{abc}\,\rho^d(x)\,\rho^e(y)\,\rho^h(x')\,U^{da}(x)\,U^{ch}(x')\,U^{be}(y)\\
(7)\quad &+K_{\alpha'}\,f^{abc}\,\rho^e(x)\,\rho^l(y)\,\rho^c(x')\,U^{de}(w)\,U^{ad}(w')\,U^{hb}(z)\,U^{hl}(y)\\
(8)\quad &-K_{\alpha'}\,f^{abc}\,\rho^d(x)\,\rho^e(y)\,\rho^h(x')\,U^{da}(w)\,U^{ch}(x')\,U^{be}(y)\\
(9)\quad &-2\,K_\beta\,f^{abc}\,\rho^e(x')\,\rho^l(y)\,\rho^c(x)\,U^{de}(x')\,U^{ad}(w)\,U^{hb}(z)\,U^{hl}(y)\\
(10)\quad &+2\,K_\beta\,f^{abc}\,\rho^e(x')\,\rho^l(y)\,\rho^c(x)\,U^{de}(w')\,U^{ad}(w)\,U^{hb}(z)\,U^{hl}(y)\\
(11)\quad &+2\,K_\beta\,f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{da}(x')\,U^{be}(y)\,U^{hc}(x)\\
(12)\quad &-2\,K_\beta\,f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{da}(w')\,U^{be}(y)\,U^{hc}(x)\\
(13)\quad &+2\,K_{\beta'}\,f^{abc}\,\rho^e(x)\,\rho^l(y)\,\rho^c(x')\,U^{de}(x)\,U^{ad}(w')\,U^{hb}(z)\,U^{hl}(y)\\
(14)\quad &-2\,K_{\beta'}\,f^{abc}\,\rho^e(x)\,\rho^l(y)\,\rho^c(x')\,U^{de}(w)\,U^{ad}(w')\,U^{hb}(z)\,U^{hl}(y)\\
(15)\quad &+2\,K_{\beta'}\,f^{abc}\,\rho^d(x)\,\rho^e(y)\,\rho^h(x')\,U^{be}(y)\,U^{ad}(w)\,U^{ch}(x')\\
(16)\quad &-2\,K_{\beta'}\,f^{abc}\,\rho^d(x)\,\rho^e(y)\,\rho^h(x')\,U^{be}(y)\,U^{ad}(x)\,U^{ch}(x')\Big]
\end{aligned}$$

Group I, Row V, part 1.

$$i\mathcal{N}\int e^{-ik\cdot(w'-w)}\Big[$$
$$\begin{aligned}
(1)\quad &-K_\beta\,f^{abc}\,\rho^d(x')\,\rho^h(x)\,\rho^e(y)\,U^{ad}(x')\,U^{be}(z)\,U^{ch}(w)\\
(2)\quad &+K_\beta\,f^{abc}\,\rho^d(x')\,\rho^h(x)\,\rho^e(y)\,U^{ad}(w')\,U^{be}(z)\,U^{ch}(w)\\
(3)\quad &+K_\beta\,f^{abc}\,\rho^d(x')\,\rho^e(x)\,\rho^h(y)\,U^{ad}(x')\,U^{ce}(x)\,U^{bh}(y)\\
(4)\quad &-K_\beta\,f^{abc}\,\rho^d(x')\,\rho^e(x)\,\rho^h(y)\,U^{ad}(w')\,U^{ce}(x)\,U^{bh}(y)\\
(5)\quad &+K_{\beta'}\,f^{abc}\,\rho^d(x)\,\rho^h(x')\,\rho^e(y)\,U^{ad}(x)\,U^{be}(z)\,U^{ch}(w')\\
(6)\quad &-K_{\beta'}\,f^{abc}\,\rho^d(x)\,\rho^h(x')\,\rho^e(y)\,U^{ad}(w)\,U^{be}(z)\,U^{ch}(w')\\
(7)\quad &+K_{\beta'}\,f^{abc}\,\rho^d(x)\,\rho^e(x')\,\rho^h(y)\,U^{ad}(w)\,U^{bh}(y)\,U^{ce}(x')\\
(8)\quad &-K_{\beta'}\,f^{abc}\,\rho^d(x)\,\rho^e(x')\,\rho^h(y)\,U^{ad}(x)\,U^{bh}(y)\,U^{ce}(x')\Big]
\end{aligned}$$

Group I, Row V, part 2.

$$i\mathcal{N}\int e^{-ik\cdot(w'-w)}\Big[$$
$$\begin{aligned}
(1)\quad &+K_\beta\,f^{abc}\,\rho^d(x')\,\rho^h(y)\,\rho^e(x)\,U^{ad}(x')\,U^{be}(w)\,U^{ch}(z)\\
(2)\quad &-K_\beta\,f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(x')\,U^{ce}(y)\,U^{bh}(x)\\
(3)\quad &-K_\beta\,f^{abc}\,\rho^d(x')\,\rho^h(y)\,\rho^e(x)\,U^{ad}(w')\,U^{be}(w)\,U^{ch}(z)\\
(4)\quad &+K_\beta\,f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(w')\,U^{ce}(y)\,U^{bh}(x)\\
(5)\quad &-K_{\beta'}\,f^{abc}\,\rho^d(x)\,\rho^h(y)\,\rho^e(x')\,U^{ad}(x)\,U^{be}(w')\,U^{ch}(z)\\
(6)\quad &+K_{\beta'}\,f^{abc}\,\rho^d(x)\,\rho^e(y)\,\rho^h(x')\,U^{ad}(x)\,U^{ce}(y)\,U^{bh}(x')\\
(7)\quad &+K_{\beta'}\,f^{abc}\,\rho^d(x)\,\rho^h(y)\,\rho^e(x')\,U^{ad}(w)\,U^{ch}(z)\,U^{be}(w')\\
(8)\quad &-K_{\beta'}\,f^{abc}\,\rho^d(x)\,\rho^e(y)\,\rho^h(x')\,U^{ad}(w)\,U^{bh}(x')\,U^{ce}(y)\Big]
\end{aligned}$$

Group II, Row I.

$$i\mathcal{N}\int e^{-ik\cdot(w'-w)}\Big[$$
$$(1)\quad -K_\alpha\,f^{abc}\,\rho^e(x')\,\rho^b(y)\,\rho^c(x)\,U^{de}(w')\,U^{ad}(w)$$

$$
\begin{aligned}
&(2)\quad +\,K_\alpha\, f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{be}(z)\,U^{ad}(w')\,U^{ch}(x)\\
&(3)\quad +\,K_\alpha\, f^{abc}\,\rho^h(x')\,\rho^l(y)\,\rho^c(x)\,U^{eh}(x')\,U^{bd}(z)\,U^{dl}(y)\,U^{ae}(w)\\
&(4)\quad -\,K_\alpha\, f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(x')\,U^{be}(y)\,U^{ch}(x)\\
&(5)\quad +2\,K_\beta\, f^{abc}\,\rho^e(x')\,\rho^b(y)\,\rho^c(x)\,U^{de}(w')\,U^{ad}(w)\\
&(6)\quad -2\,K_\beta\, f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(w')\,U^{eb}(z)\,U^{ch}(x)\\
&(7)\quad -2\,K_\beta\, f^{abc}\,\rho^h(x')\,\rho^l(y)\,\rho^c(x)\,U^{eh}(x')\,U^{bd}(z)\,U^{dl}(y)\,U^{ea}(w)\\
&(8)\quad +2\,K_\beta\, f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{da}(x')\,U^{be}(y)\,U^{ch}(x)\\
&(9)\quad +\,K_{\alpha'}\, f^{abc}\,\rho^c(x')\,\rho^e(x)\,\rho^b(y)\,U^{ad}(w')\,U^{de}(w)\\
&(10)\quad -\,K_{\alpha'}\, f^{abc}\,\rho^c(x')\,\rho^h(x)\,\rho^l(y)\,U^{ad}(w')\,U^{be}(z)\,U^{dh}(x)\,U^{el}(y)\\
&(11)\quad -\,K_{\alpha'}\, f^{abc}\,\rho^e(x')\,\rho^h(x)\,\rho^d(y)\,U^{bd}(z)\,U^{ce}(x')\,U^{ah}(w)\\
&(12)\quad +\,K_{\alpha'}\, f^{abc}\,\rho^d(x')\,\rho^e(x)\,\rho^h(y)\,U^{cd}(x')\,U^{ae}(x)\,U^{bh}(y)\\
&(13)\quad -2\,K_{\beta'}\, f^{abc}\,\rho^c(x')\,\rho^e(x)\,\rho^b(y)\,U^{ad}(w')\,U^{de}(w)\\
&(14)\quad +2\,K_{\beta'}\, f^{abc}\,\rho^c(x')\,\rho^h(x)\,\rho^l(y)\,U^{ad}(w')\,U^{be}(z)\,U^{dh}(x)\,U^{el}(y)\\
&(15)\quad +2\,K_{\beta'}\, f^{abc}\,\rho^e(x')\,\rho^h(x)\,\rho^d(y)\,U^{ce}(x')\,U^{db}(z)\,U^{ah}(w)\\
&(16)\quad -2\,K_{\beta'}\, f^{abc}\,\rho^d(x')\,\rho^e(x)\,\rho^h(y)\,U^{cd}(x')\,U^{ae}(x)\,U^{bh}(y)\Big]
\end{aligned}
$$

Group III, Row I.

$$
\begin{aligned}
&i\mathcal{N}\int e^{-ik\cdot(w'-w)}\Big[\\
&(1)\quad -\,K_\alpha\, f^{abc}\,\rho^e(x')\,\rho^l(y)\,\rho^c(x)\,U^{de}(x')\,U^{hl}(y)\,U^{hb}(z)\,U^{ad}(w)\\
&(2)\quad +\,K_\alpha\, f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(x')\,U^{be}(y)\,U^{ch}(x)\\
&(3)\quad +\,K_\alpha\, f^{abc}\,\rho^l(x')\,\rho^e(y)\,\rho^c(x)\,U^{bd}(z)\,U^{de}(y)\,U^{hl}(w')\,U^{ah}(w)\\
&(4)\quad -\,K_\alpha\, f^{abc}\,\rho^e(x')\,\rho^d(y)\,\rho^h(x)\,U^{ae}(w')\,U^{bd}(y)\,U^{ch}(x)\\
&(5)\quad +2\,K_\beta\, f^{abc}\,\rho^e(x')\,\rho^l(y)\,\rho^c(x)\,U^{de}(x')\,U^{hl}(y)\,U^{hb}(z)\,U^{ad}(w)\\
&(6)\quad -2\,K_\beta\, f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(x')\,U^{be}(y)\,U^{ch}(x)\\
&(7)\quad -2\,K_\beta\, f^{abc}\,\rho^l(x')\,\rho^e(y)\,\rho^c(x)\,U^{bd}(z)\,U^{de}(y)\,U^{hl}(w')\,U^{ah}(w)\\
&(8)\quad +2\,K_\beta\, f^{abc}\,\rho^e(x')\,\rho^d(y)\,\rho^h(x)\,U^{ae}(w')\,U^{bd}(y)\,U^{ch}(x)\\
&(9)\quad +\,K_{\alpha'}\, f^{abc}\,\rho^e(x)\,\rho^l(y)\,\rho^c(x')\,U^{de}(x)\,U^{hl}(y)\,U^{hb}(z)\,U^{ad}(w')\\
&(10)\quad -\,K_{\alpha'}\, f^{abc}\,\rho^d(x)\,\rho^e(y)\,\rho^h(x')\,U^{ad}(x)\,U^{be}(y)\,U^{ch}(x')\\
&(11)\quad -\,K_{\alpha'}\, f^{abc}\,\rho^l(x)\,\rho^e(y)\,\rho^c(x')\,U^{bd}(z)\,U^{de}(y)\,U^{hl}(w)\,U^{ha}(w')\\
&(12)\quad +\,K_{\alpha'}\, f^{abc}\,\rho^e(x)\,\rho^d(y)\,\rho^h(x')\,U^{bd}(y)\,U^{ae}(w)\,U^{ch}(x')\\
&(13)\quad -2\,K_{\beta'}\, f^{abc}\,\rho^e(x)\,\rho^l(y)\,\rho^c(x')\,U^{de}(x)\,U^{hl}(y)\,U^{hb}(z)\,U^{ad}(w')\\
&(14)\quad +2\,K_{\beta'}\, f^{abc}\,\rho^d(x)\,\rho^e(y)\,\rho^h(x')\,U^{ad}(x)\,U^{be}(y)\,U^{ch}(x')\\
&(15)\quad +2\,K_{\beta'}\, f^{abc}\,\rho^l(x)\,\rho^e(y)\,\rho^c(x')\,U^{de}(y)\,U^{hl}(w)\,U^{db}(z)\,U^{ha}(w')\ \text{[T1]}\\
&(16)\quad -2\,K_{\beta'}\, f^{abc}\,\rho^e(x)\,\rho^d(y)\,\rho^h(x')\,U^{bd}(y)\,U^{ae}(w)\,U^{ch}(x')\Big]
\end{aligned}
$$

Group III, Row II.

$$
\begin{aligned}
&i\mathcal{N}\int e^{-ik\cdot(w'-w)}\Big[\\
&(1)\quad -\,K_\alpha\, f^{abc}\,\rho^e(y)\,\rho^l(x')\,\rho^c(x)\,U^{de}(y)\,U^{hl}(x')\,U^{bd}(z)\,U^{ah}(w)\\
&(2)\quad +\,K_\alpha\, f^{abc}\,\rho^d(y)\,\rho^e(x')\,\rho^h(x)\,U^{bd}(y)\,U^{ae}(x')\,U^{ch}(x)\\
&(3)\quad +\,K_\alpha\, f^{abc}\,\rho^b(y)\,\rho^e(x')\,\rho^c(x)\,U^{de}(x')\,U^{ad}(w)\\
&(4)\quad -\,K_\alpha\, f^{abc}\,\rho^e(y)\,\rho^d(x')\,\rho^h(x)\,U^{ad}(x')\,U^{eb}(z)\,U^{ch}(x)\\
&(5)\quad +2\,K_\beta\, f^{abc}\,\rho^e(y)\,\rho^l(x')\,\rho^c(x)\,U^{de}(y)\,U^{hl}(x')\,U^{bd}(z)\,U^{ha}(w)\\
&(6)\quad -2\,K_\beta\, f^{abc}\,\rho^d(y)\,\rho^e(x')\,\rho^h(x)\,U^{db}(y)\,U^{ae}(x')\,U^{ch}(x)\\
&(7)\quad -2\,K_\beta\, f^{abc}\,\rho^b(y)\,\rho^e(x')\,\rho^c(x)\,U^{de}(x')\,U^{da}(w)\\
&(8)\quad +2\,K_\beta\, f^{abc}\,\rho^e(y)\,\rho^d(x')\,\rho^h(x)\,U^{ad}(x')\,U^{eb}(z)\,U^{ch}(x)\\
&(9)\quad +\,K_{\alpha'}\, f^{abc}\,\rho^e(y)\,\rho^l(x)\,\rho^c(x')\,U^{de}(y)\,U^{hl}(x)\,U^{bd}(z)\,U^{ah}(w')\\
&(10)\quad -\,K_{\alpha'}\, f^{abc}\,\rho^d(y)\,\rho^e(x)\,\rho^h(x')\,U^{bd}(y)\,U^{ae}(x)\,U^{ch}(x')\\
&(11)\quad -\,K_{\alpha'}\, f^{abc}\,\rho^b(y)\,\rho^e(x)\,\rho^c(x')\,U^{de}(x)\,U^{ad}(w')
\end{aligned}
$$

$$
\begin{aligned}
(12)\quad & + K_{\alpha'}\, f^{abc}\, \rho^{e}(y)\, \rho^{d}(x)\, \rho^{h}(x')\, U^{ad}(x)\, U^{eb}(z)\, U^{ch}(x') \\
(13)\quad & -2\, K_{\beta'}\, f^{abc}\, \rho^{e}(y)\, \rho^{l}(x)\, \rho^{c}(x')\, U^{de}(y)\, U^{hl}(x)\, U^{bd}(z)\, U^{ha}(w') \ \text{[T2]} \\
(14)\quad & + 2\, K_{\beta'}\, f^{abc}\, \rho^{d}(y)\, \rho^{e}(x)\, \rho^{h}(x')\, U^{bd}(y)\, U^{ae}(x)\, U^{ch}(x') \\
(15)\quad & + 2\, K_{\beta'}\, f^{abc}\, \rho^{b}(y)\, \rho^{e}(x)\, \rho^{c}(x')\, U^{de}(x)\, U^{ad}(w') \\
(16)\quad & - 2\, K_{\beta'}\, f^{abc}\, \rho^{e}(y)\, \rho^{d}(x)\, \rho^{h}(x')\, U^{ad}(x)\, U^{eb}(z)\, U^{ch}(x') \Big]
\end{aligned}
$$

Group III, Row III.

$$
\begin{aligned}
& i\mathcal{N} \int e^{-ik\cdot(w'-w)} \Big[\\
(1)\quad & + 2\, K_{\beta}\, f^{abc}\, \rho^{d}(x)\, \rho^{h}(x')\, \rho^{e}(y)\, U^{ad}(w)\, U^{be}(z)\, U^{ch}(w') \\
(2)\quad & - 2\, K_{\beta}\, f^{abc}\, \rho^{d}(x)\, \rho^{e}(x')\, \rho^{h}(y)\, U^{ad}(w)\, U^{ce}(x')\, U^{bh}(y) \\
(3)\quad & - 2\, K_{\beta}\, f^{abc}\, \rho^{e}(x)\, \rho^{h}(x')\, \rho^{d}(y)\, U^{db}(z)\, U^{ae}(x)\, U^{ch}(w') \\
(4)\quad & + 2\, K_{\beta}\, f^{abc}\, \rho^{d}(x)\, \rho^{e}(x')\, \rho^{h}(y)\, U^{ad}(x)\, U^{ce}(x')\, U^{bh}(y) \\
(5)\quad & - 2\, K_{\beta'}\, f^{abc}\, \rho^{d}(x')\, \rho^{h}(x)\, \rho^{e}(y)\, U^{ad}(w')\, U^{be}(z)\, U^{ch}(w) \\
(6)\quad & + 2\, K_{\beta'}\, f^{abc}\, \rho^{d}(x')\, \rho^{e}(x)\, \rho^{h}(y)\, U^{ad}(w')\, U^{ce}(x)\, U^{bh}(y) \\
(7)\quad & + 2\, K_{\beta'}\, f^{abc}\, \rho^{e}(x')\, \rho^{h}(x)\, \rho^{d}(y)\, U^{bd}(z)\, U^{ae}(x')\, U^{ch}(w) \\
(8)\quad & - 2\, K_{\beta'}\, f^{abc}\, \rho^{d}(x')\, \rho^{e}(x)\, \rho^{h}(y)\, U^{ad}(x')\, U^{ce}(x)\, U^{bh}(y) \Big]
\end{aligned}
$$

Group III, Row V.

$$
\begin{aligned}
& i\mathcal{N} \int e^{-ik\cdot(w'-w)} \Big[\\
(1)\quad & + 2\, K_{\beta}\, f^{abc}\, \rho^{d}(x)\, \rho^{e}(y)\, \rho^{h}(x')\, U^{ad}(w)\, U^{be}(y)\, U^{ch}(x') \\
(2)\quad & - 2\, K_{\beta}\, f^{abc}\, \rho^{d}(x)\, \rho^{h}(y)\, \rho^{e}(x')\, U^{ad}(w)\, U^{ce}(w')\, U^{bh}(y) \\
(3)\quad & - 2\, K_{\beta}\, f^{abc}\, \rho^{d}(x)\, \rho^{e}(y)\, \rho^{h}(x')\, U^{be}(y)\, U^{ad}(x)\, U^{ch}(x') \\
(4)\quad & + 2\, K_{\beta}\, f^{abc}\, \rho^{d}(x)\, \rho^{h}(y)\, \rho^{e}(x')\, U^{ad}(x)\, U^{ce}(w')\, U^{bh}(y) \\
(5)\quad & - 2\, K_{\beta'}\, f^{abc}\, \rho^{d}(x')\, \rho^{e}(y)\, \rho^{h}(x)\, U^{ad}(w')\, U^{be}(y)\, U^{ch}(x) \\
(6)\quad & + 2\, K_{\beta'}\, f^{abc}\, \rho^{d}(x')\, \rho^{h}(y)\, \rho^{e}(x)\, U^{ad}(w')\, U^{ce}(w)\, U^{bh}(y) \\
(7)\quad & + 2\, K_{\beta'}\, f^{abc}\, \rho^{d}(x')\, \rho^{e}(y)\, \rho^{h}(x)\, U^{be}(y)\, U^{ad}(x')\, U^{ch}(x) \\
(8)\quad & - 2\, K_{\beta'}\, f^{abc}\, \rho^{d}(x')\, \rho^{h}(y)\, \rho^{e}(x)\, U^{ad}(x')\, U^{ce}(w)\, U^{bh}(y) \Big]
\end{aligned}
$$

Group III, Row VI.

$$
\begin{aligned}
& i\mathcal{N} \int e^{-ik\cdot(w'-w)} \Big[\\
(1)\quad & + 2\, K_{\beta}\, f^{abc}\, \rho^{d}(x)\, \rho^{e}(x')\, \rho^{h}(y)\, U^{ad}(w)\, U^{ce}(x')\, U^{bh}(y) \\
(2)\quad & - 2\, K_{\beta}\, f^{abc}\, \rho^{d}(x)\, \rho^{h}(x')\, \rho^{e}(y)\, U^{ad}(w)\, U^{be}(z)\, U^{ch}(x') \\
(3)\quad & - 2\, K_{\beta}\, f^{abc}\, \rho^{d}(x)\, \rho^{e}(x')\, \rho^{h}(y)\, U^{bh}(y)\, U^{ad}(x)\, U^{ce}(x') \\
(4)\quad & + 2\, K_{\beta}\, f^{abc}\, \rho^{e}(x)\, \rho^{h}(x')\, \rho^{d}(y)\, U^{ae}(x)\, U^{ch}(x')\, U^{db}(z) \\
(5)\quad & - 2\, K_{\beta'}\, f^{abc}\, \rho^{d}(x')\, \rho^{e}(x)\, \rho^{h}(y)\, U^{ad}(w')\, U^{bh}(y)\, U^{ce}(x) \\
(6)\quad & + 2\, K_{\beta'}\, f^{abc}\, \rho^{d}(x')\, \rho^{h}(x)\, \rho^{e}(y)\, U^{ad}(w')\, U^{be}(z)\, U^{ch}(x) \\
(7)\quad & + 2\, K_{\beta'}\, f^{abc}\, \rho^{d}(x')\, \rho^{e}(x)\, \rho^{h}(y)\, U^{bh}(y)\, U^{ad}(x')\, U^{ce}(x) \\
(8)\quad & - 2\, K_{\beta'}\, f^{abc}\, \rho^{e}(x')\, \rho^{h}(x)\, \rho^{d}(y)\, U^{ae}(x')\, U^{ch}(x)\, U^{bd}(z) \Big]
\end{aligned}
$$

5 The cancellation, network by network

Every symmetrized term is written in common variables and simplified with $U^{bc} = U^{cb}$, $U(p)U(p) = 1$ and $f^{acd}f^{bcd} = N_c\delta^{ab}$. Within a kernel structure, the colour structures that are equal up to sign form a network; the coefficients of each network are added exactly. Each table lists, for every network:

- the contribution of every row (in units of $i\mathcal{N}$ times the network);
- the sum as written (T1–T3 fixed; red rows included, $\mathcal{D}$ excluded);
- the sum after F3 and F4 (red rows excluded, $\mathcal{D}$ included).

$\mathrm{I.II_{F4}}$ and $\mathrm{I.III_{F3}}$ are the removed terms, and $\mathcal{D}$ the added ones. As written, 20 of the 48 networks do not cancel. After the fixes all 40 remaining networks cancel.

Kernel structure 1: $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)$

colour network	contributions (row: coefficient)	as written	fixed
$f^{abc}\,U^{de}(w')\,U^{eb}(w)\,\rho^d(x')\,\rho^a(y)\,\rho^c(x)$	A,B: $-i$; II.I: i; $\mathrm{I.II_{F4}}$: $-2i$	$-2i$	0 ✓
$f^{abc}\,U^{db}(w)\,U^{ed}(x')\,\rho^e(x')\,\rho^a(y)\,\rho^c(x)$	A,B: i; III.II: $-i$; $\mathrm{I.II_{F4}}$: $2i$	$2i$	0 ✓
$f^{abc}\,U^{db}(w')\,U^{ec}(x)\,U^{ah}(y)\,\rho^d(x')\,\rho^h(y)\,\rho^e(x)$	A,B: $-i$; I.II: $2i$; I.III: $-i$; I.IV: $-i$; III.I: i	0	0 ✓
$f^{abc}\,U^{dc}(x)\,U^{be}(x')\,U^{ah}(y)\,\rho^e(x')\,\rho^h(y)\,\rho^d(x)$	A,B: i; I.II: $-2i$; I.III: i; I.IV: i; II.I: i; III.I: $-i$; III.II: $-i$	0	0 ✓
$f^{abc}\,U^{ad}(x')\,U^{ce}(x)\,U^{bh}(z)\,\rho^d(x')\,\rho^e(x)\,\rho^h(y)$	I.III: i; III.II: $-i$	0	0 ✓
$f^{abc}\,U^{ad}(w')\,U^{ce}(x)\,U^{bh}(z)\,\rho^d(x')\,\rho^e(x)\,\rho^h(y)$	I.III: $-i$; II.I: i	0	0 ✓
$f^{abc}\,U^{de}(x')\,U^{ad}(w)\,U^{hb}(z)\,U^{hl}(y)\,\rho^e(x')$ $\rho^l(y)\,\rho^c(x)$	I.IV: i; II.I: i; III.I: $-i$; III.II: $-i$	0	0 ✓
$f^{abc}\,U^{de}(w')\,U^{ad}(w)\,U^{hb}(z)\,U^{hl}(y)\,\rho^e(x')$ $\rho^l(y)\,\rho^c(x)$	I.IV: $-i$; III.I: i	0	0 ✓

Kernel structure 2: $\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')$

colour network	contributions (row: coefficient)	as written	fixed
$f^{abc}\,U^{db}(w')\,U^{ed}(x)\,\rho^c(x')\,\rho^a(y)\,\rho^e(x)$	A,B: $-i$; III.II: i; $\mathrm{I.II_{F4}}$: $-2i$	$-2i$	0 ✓
$f^{abc}\,U^{bd}(w')\,U^{ed}(w)\,\rho^c(x')\,\rho^a(y)\,\rho^e(x)$	A,B: i; II.I: $-i$; $\mathrm{I.II_{F4}}$: $2i$	$2i$	0 ✓
$f^{abc}\,U^{dc}(x')\,U^{eb}(w)\,U^{ah}(y)\,\rho^d(x')\,\rho^e(x)\,\rho^h(y)$	A,B: i; I.II: $-2i$; I.III: i; I.IV: i; III.I: $-i$	0	0 ✓
$f^{abc}\,U^{dc}(x')\,U^{be}(x)\,U^{ah}(y)\,\rho^d(x')\,\rho^e(x)\,\rho^h(y)$	A,B: $-i$; I.II: $2i$; I.III: $-i$; I.IV: $-i$; II.I: $-i$; III.I: i; III.II: i	0	0 ✓
$f^{abc}\,U^{ad}(x)\,U^{ce}(x')\,U^{bh}(z)\,\rho^d(x)\,\rho^e(x')\,\rho^h(y)$	I.III: $-i$; III.II: i	0	0 ✓
$f^{abc}\,U^{ad}(w)\,U^{ce}(x')\,U^{bh}(z)\,\rho^d(x)\,\rho^e(x')\,\rho^h(y)$	I.III: i; II.I: $-i$	0	0 ✓
$f^{abc}\,U^{de}(x)\,U^{ad}(w')\,U^{hb}(z)\,U^{hl}(y)\,\rho^e(x)$ $\rho^l(y)\,\rho^c(x')$	I.IV: $-i$; II.I: $-i$; III.I: i; III.II: i	0	0 ✓
$f^{abc}\,U^{de}(w)\,U^{ad}(w')\,U^{hb}(z)\,U^{hl}(y)\,\rho^e(x)$ $\rho^l(y)\,\rho^c(x')$	I.IV: i; III.I: $-i$	0	0 ✓

Kernel structure 3: $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(y-z)\cdot\mathcal{K}(w-z)$

colour network	contributions (row: coefficient)	as written	fixed
$f^{abc}\,U^{de}(x')\,U^{ad}(w)\,\rho^e(x')\,\rho^b(x)\,\rho^c(y)$	I.II: $-2i$; III.II: $2i$	0	0 ✓
$f^{abc}\,U^{ad}(x')\,U^{be}(x)\,U^{ch}(y)\,\rho^d(x')\,\rho^e(x)\,\rho^h(y)$	I.II: $2i$; $\mathcal{D}$: $2i$; I.IV: $-2i$; I.V: $-2i$; II.I: $-2i$; III.I: $2i$; III.II: $2i$; III.III: $2i$; III.V: $-2i$; III.VI: $-2i$	$-2i$	0 ✓
$f^{abc}\,U^{de}(w')\,U^{ad}(w)\,\rho^e(x')\,\rho^b(x)\,\rho^c(y)$	I.II: $2i$; II.I: $-2i$	0	0 ✓
$f^{abc}\,U^{ad}(w')\,U^{be}(x)\,U^{ch}(y)\,\rho^d(x')\,\rho^e(x)\,\rho^h(y)$	I.II: $-2i$; $\mathcal{D}$: $-2i$; I.IV: $2i$; I.V: $2i$; III.I: $-2i$; III.V: $2i$	$2i$	0 ✓
$f^{abc}\,U^{de}(x')\,U^{ad}(w)\,U^{hb}(z)\,U^{hl}(y)\,\rho^e(x')$ $\rho^l(y)\,\rho^c(x)$	I.IV: $-2i$; II.I: $-2i$; III.I: $2i$; III.II: $2i$	0	0 ✓
$f^{abc}\,U^{de}(w')\,U^{ad}(w)\,U^{hb}(z)\,U^{hl}(y)\,\rho^e(x')$ $\rho^l(y)\,\rho^c(x)$	I.IV: $2i$; III.I: $-2i$	0	0 ✓
$f^{abc}\,U^{ad}(x')\,U^{be}(z)\,U^{ch}(w)\,\rho^d(x')\,\rho^h(x)\,\rho^e(y)$	I.V: $-2i$; III.VI: $2i$	0	0 ✓
$f^{abc}\,U^{ad}(w')\,U^{be}(z)\,U^{ch}(w)\,\rho^d(x')\,\rho^h(x)$ $\rho^e(y)$	I.V: $2i$; III.III: $-2i$	0	0 ✓
$f^{abc}\,U^{ad}(w')\,U^{eb}(z)\,U^{ch}(x)\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)$	II.I: $-2i$; III.III: $2i$	0	0 ✓
$f^{abc}\,U^{ad}(x')\,U^{eb}(z)\,U^{ch}(x)\,\rho^e(y)\,\rho^d(x')\,\rho^h(x)$	III.II: $2i$; III.VI: $-2i$	0	0 ✓
$f^{abc}\,U^{ad}(w)\,U^{ce}(x')\,U^{bh}(y)\,\rho^d(x)\,\rho^e(x')\,\rho^h(y)$	$\mathcal{D}$: $-2i$; III.III: $-2i$; III.V: $2i$; III.VI: $2i$	$2i$	0 ✓
$f^{abc}\,U^{ad}(w)\,U^{ce}(w')\,U^{bh}(y)\,\rho^d(x)\,\rho^h(y)$ $\rho^e(x')$	$\mathcal{D}$: $2i$; III.V: $-2i$	$-2i$	0 ✓

Kernel structure 4: $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(y-z)\cdot\mathcal{K}(w'-z)$

colour network	contributions (row: coefficient)	as written	fixed
$f^{abc}\,U^{de}(x)\,U^{ad}(w')\,\rho^e(x)\,\rho^b(x')\,\rho^c(y)$	I.II: $2i$; III.II: $-2i$	0	0 ✓

colour network	contributions (row: coefficient)	as written	fixed
$f^{abc}\,U^{ad}(x)\,U^{be}(x')\,U^{ch}(y)\,\rho^{d}(x)\,\rho^{e}(x')\,\rho^{h}(y)$	I.II: $-2i$; $\mathcal{D}$: $-2i$; I.IV: $2i$; I.V: $2i$; II.I: $2i$; III.I: $-2i$; III.II: $-2i$; III.III: $-2i$; III.V: $2i$; III.VI: $2i$	$2i$	0 ✓
$f^{abc}\,U^{de}(w)\,U^{ad}(w')\,\rho^{e}(x)\,\rho^{b}(x')\,\rho^{c}(y)$	I.II: $-2i$; II.I: $2i$	0	0 ✓
$f^{abc}\,U^{ad}(w)\,U^{be}(x')\,U^{ch}(y)\,\rho^{d}(x)\,\rho^{e}(x')\,\rho^{h}(y)$	I.II: $2i$; $\mathcal{D}$: $2i$; I.IV: $-2i$; I.V: $-2i$; III.I: $2i$; III.V: $-2i$	$-2i$	0 ✓
$f^{abc}\,U^{de}(x)\,U^{ad}(w')\,U^{hb}(z)\,U^{hl}(y)\,\rho^{e}(x)\,\rho^{l}(y)\,\rho^{c}(x')$	I.IV: $2i$; II.I: $2i$; III.I: $-2i$; III.II: $-2i$	0	0 ✓
$f^{abc}\,U^{de}(w)\,U^{ad}(w')\,U^{hb}(z)\,U^{hl}(y)\,\rho^{e}(x)\,\rho^{l}(y)\,\rho^{c}(x')$	I.IV: $-2i$; III.I: $2i$	0	0 ✓
$f^{abc}\,U^{ad}(x)\,U^{be}(z)\,U^{ch}(w')\,\rho^{d}(x)\,\rho^{h}(x')\,\rho^{e}(y)$	I.V: $2i$; III.VI: $-2i$	0	0 ✓
$f^{abc}\,U^{ad}(w)\,U^{be}(z)\,U^{ch}(w')\,\rho^{d}(x)\,\rho^{h}(x')\,\rho^{e}(y)$	I.V: $-2i$; III.III: $2i$	0	0 ✓
$f^{abc}\,U^{cd}(x')\,U^{eb}(z)\,U^{ah}(w)\,\rho^{d}(x')\,\rho^{h}(x)\,\rho^{e}(y)$	II.I: $2i$; III.III: $-2i$	0	0 ✓
$f^{abc}\,U^{ad}(x)\,U^{eb}(z)\,U^{ch}(x')\,\rho^{e}(y)\,\rho^{d}(x)\,\rho^{h}(x')$	III.II: $-2i$; III.VI: $2i$	0	0 ✓
$f^{abc}\,U^{ad}(w')\,U^{ce}(x)\,U^{bh}(y)\,\rho^{d}(x')\,\rho^{e}(x)\,\rho^{h}(y)$	$\mathcal{D}$: $2i$; III.III: $2i$; III.V: $-2i$; III.VI: $-2i$	$-2i$	0 ✓
$f^{abc}\,U^{ad}(w')\,U^{ce}(w)\,U^{bh}(y)\,\rho^{d}(x')\,\rho^{h}(y)\,\rho^{e}(x)$	$\mathcal{D}$: $-2i$; III.V: $2i$	$2i$	0 ✓

Kernel structure 5: $\mathcal{K}(x-z)\cdot\mathcal{K}(x'-w')\;\phi_w(y)$

colour network	contributions (row: coefficient)	as written	fixed
$f^{abc}\,U^{ad}(x')\,U^{ce}(y)\,U^{bh}(z)\,\rho^{d}(x')\,\rho^{e}(y)\,\rho^{h}(x)$	$\mathrm{I.III_{F3}}$: $2i$	$2i$	–
$f^{abc}\,U^{ad}(x')\,U^{ce}(y)\,U^{bh}(x)\,\rho^{d}(x')\,\rho^{e}(y)\,\rho^{h}(x)$	$\mathrm{I.III_{F3}}$: $-2i$	$-2i$	–
$f^{abc}\,U^{ad}(w')\,U^{ce}(y)\,U^{bh}(z)\,\rho^{d}(x')\,\rho^{e}(y)\,\rho^{h}(x)$	$\mathrm{I.III_{F3}}$: $-2i$	$-2i$	–
$f^{abc}\,U^{ad}(w')\,U^{ce}(y)\,U^{bh}(x)\,\rho^{d}(x')\,\rho^{e}(y)\,\rho^{h}(x)$	$\mathrm{I.III_{F3}}$: $2i$	$2i$	–

Kernel structure 6: $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-z)\;\phi_{w'}(y)$

colour network	contributions (row: coefficient)	as written	fixed
$f^{abc}\,U^{ad}(x)\,U^{ce}(y)\,U^{bh}(z)\,\rho^{d}(x)\,\rho^{e}(y)\,\rho^{h}(x')$	$\mathrm{I.III_{F3}}$: $-2i$	$-2i$	–
$f^{abc}\,U^{ad}(x)\,U^{ce}(y)\,U^{bh}(x')\,\rho^{d}(x)\,\rho^{e}(y)\,\rho^{h}(x')$	$\mathrm{I.III_{F3}}$: $2i$	$2i$	–
$f^{abc}\,U^{ad}(w)\,U^{ce}(y)\,U^{bh}(z)\,\rho^{d}(x)\,\rho^{e}(y)\,\rho^{h}(x')$	$\mathrm{I.III_{F3}}$: $2i$	$2i$	–
$f^{abc}\,U^{ad}(w)\,U^{ce}(y)\,U^{bh}(x')\,\rho^{d}(x)\,\rho^{e}(y)\,\rho^{h}(x')$	$\mathrm{I.III_{F3}}$: $-2i$	$-2i$	–

6 The operator order

6.1 Why the order matters for the two-ρ terms

- *c.c.* The cross section is the expectation value of a Hermitian operator. The "+c.c." of the notes is the Hermitian conjugate. With $\rho^\dagger=\rho$ and real U, f,

$$\big(c\,ABC\big)^\dagger = c^*\,CBA, \qquad CBA = S(ABC) - \tfrac{1}{4}\big(\{[A,B],C\}+\{[A,C],B\}+\{[B,C],A\}\big)+\ldots \tag{6.1}$$

  So the c.c. has the same symmetrized part as c^*ABC, but two-ρ terms of the opposite sign. In your list every c.c. term keeps the order of the original term.

- *Anticommutators.* Group I Row V, Group II Row I and Group III Row III (part 1) contain $\rho\{\rho,\rho\}$ or $\{\rho,\rho\}\rho$ in the notes. Your list writes each as twice one ordering, with twice the coefficient. Then

$$\{B,C\}A - 2\,BCA = -[B,C]A = -\tfrac{1}{2}\{[B,C],A\} + (\text{one }\rho). \tag{6.2}$$

  The symmetrized parts agree, but the two-ρ terms differ.

- *Group I Row II* is not affected by the second point: $f^{abc}\big(\rho^b(x)\rho^c(y)-\rho^b(y)\rho^c(x)\big)=f^{abc}\{\rho^b(x),\rho^c(y)\}$ exactly, so your two orderings are the anticommutator.

The rules, row by row (common variables):

row	amplitude (α,β,γ)	c.c. (α',β',γ')
four-ρ remainders A, B	as written (ABC)	– (no c.c.)
Group I Row II	$\rho(x')\{\rho(x),\rho(y)\}$ (as written)	$\{\rho(x'),\rho(y)\}\rho(x)$ (reversed)
Group I Row III, $\mathcal{D}$	as written (ABC)	reversed (CBA)
Group I Row IV	as written	reversed
Group I Row V	$A\{B,C\}$, half the coefficient	$\{B,C\}A$, half the coefficient
Group II Row I	$\{B,C\}A$ (α,β), $A\{B,C\}$ (α',β'), half the coefficient	– (the two are conjugates)
Group III Rows I, II, V, VI	as written	reversed
Group III Row III	$A\{B,C\}$, half the coefficient	$\{B,C\}A$, half the coefficient

6.2 Literal order versus operator order

Table 2 compares the two-ρ terms of each source obtained by reordering your list literally with those of the operator order. Each value is pointwise at one random configuration.

source	literal order	operator order	
A,B	-0.04674	-0.04674	same
I.II$^{3\rho}$	0.00813	0.01708	different
I.III$^{3\rho}$	0.01140	0.02487	different
I.IV$^{3\rho}$	0.00174	0.01444	different
I.V$^{3\rho}$	-0.10340	0.02815	different
II.I$^{3\rho}$	-0.08511	-0.08606	different
III.I$^{3\rho}$	-0.00174	-0.01444	different
III.II$^{3\rho}$	-0.00890	0.06102	different
III.III$^{3\rho}$	0.22008	-0.05772	different
III.V$^{3\rho}$	-0.00579	0.03489	different
III.VI$^{3\rho}$	-0.00262	-0.04975	different
$\mathcal{D}$	-0.00579	-0.03489	different

seed	total, literal	total, operator	JIMWLK⊗LO
3	-0.089554	-0.179957	-0.179957
7	-358.895453	-515.390339	-515.390339
11	3.560998	3.400970	3.400970

Table 2: Left: two-ρ content of each source of three-ρ terms (after the fixes). Right: total two-ρ terms (these sources plus the genuine two-ρ terms and the lists of `4rho.tex`) against JIMWLK⊗LO. Only the operator order gives JIMWLK⊗LO.

6.3 The corrected list in operator order

Your terms after the fixes, with your colour indices, in operator order. Terms whose order or form changed are blue. For an anticommutator the coefficient is half of yours: your single ordering counted it twice. Group I Row II keeps your two orderings per pair, which are exactly $\rho\{\rho,\rho\}$ by the f^{abc} antisymmetry; in its c.c. (part 2) each line is reversed.

From four ρ's, d^3N_a.

$$
\begin{aligned}
&-i\mathcal{N}\int e^{-ik\cdot(w'-w)}\Big[\\
&(1)\quad +K_\alpha\, f^{abc}\,\rho^d(x')\,\rho^a(y)\,\rho^c(x)\,U^{de}(w')\,U^{eb}(w)\\
&(2)\quad -K_\alpha\, f^{abc}\,\rho^e(x')\,\rho^a(y)\,\rho^c(x)\,U^{db}(w)\,U^{ed}(x')\\
&(3)\quad +K_{\alpha'}\, f^{abc}\,\rho^c(x')\,\rho^a(y)\,\rho^e(x)\,U^{db}(w')\,U^{ed}(x)\\
&(4)\quad -K_{\alpha'}\, f^{abc}\,\rho^c(x')\,\rho^a(y)\,\rho^e(x)\,U^{bd}(w')\,U^{ed}(w)\Big]
\end{aligned}
$$

From four ρ's, d^3N_b.

$$
\begin{aligned}
&i\mathcal{N}\int e^{-ik\cdot(w'-w)}\Big[\\
&(1)\quad -K_\alpha\, f^{abc}\,\rho^d(x')\,\rho^h(y)\,\rho^e(x)\,U^{db}(w')\,U^{ec}(x)\,U^{ah}(y)\\
&(2)\quad +K_{\alpha'}\, f^{abc}\,\rho^d(x')\,\rho^e(x)\,\rho^h(y)\,U^{dc}(x')\,U^{eb}(w)\,U^{ah}(y)\\
&(3)\quad -K_{\alpha'}\, f^{abc}\,\rho^d(x')\,\rho^e(x)\,\rho^h(y)\,U^{dc}(x')\,U^{be}(x)\,U^{ah}(y)\\
&(4)\quad +K_\alpha\, f^{abc}\,\rho^e(x')\,\rho^h(y)\,\rho^d(x)\,U^{dc}(x)\,U^{be}(x')\,U^{ah}(y)\Big]
\end{aligned}
$$

Group I, Row II, part 1.

$$
\begin{aligned}
&i\mathcal{N}\int e^{-ik\cdot(w'-w)}\Big[\\
&(3)\quad -K_\alpha\, f^{abc}\,\rho^d(x')\,\rho^e(x)\,\rho^h(y)\,U^{ad}(x')\,U^{be}(x)\,U^{ch}(y)\\
&(4)\quad +K_\alpha\, f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(x')\,U^{be}(y)\,U^{ch}(x)\\
&(7)\quad +K_\alpha\, f^{abc}\,\rho^d(x')\,\rho^e(x)\,\rho^h(y)\,U^{ad}(w')\,U^{be}(x)\,U^{ch}(y)
\end{aligned}
$$

$$
\begin{aligned}
(8)\quad & -\,K_\alpha\, f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(w')\,U^{be}(y)\,U^{ch}(x)\\
(9)\quad & -\,K_\beta\, f^{abc}\,\rho^e(x')\,\rho^b(x)\,\rho^c(y)\,U^{de}(x')\,U^{ad}(w)\\
(10)\quad & +\,K_\beta\, f^{abc}\,\rho^e(x')\,\rho^b(y)\,\rho^c(x)\,U^{de}(x')\,U^{ad}(w)\\
(11)\quad & +\,K_\beta\, f^{abc}\,\rho^d(x')\,\rho^e(x)\,\rho^h(y)\,U^{ad}(x')\,U^{be}(x)\,U^{ch}(y)\\
(12)\quad & -\,K_\beta\, f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(x')\,U^{be}(y)\,U^{ch}(x)\\
(13)\quad & +\,K_\beta\, f^{abc}\,\rho^e(x')\,\rho^b(x)\,\rho^c(y)\,U^{de}(w')\,U^{ad}(w)\\
(14)\quad & -\,K_\beta\, f^{abc}\,\rho^e(x')\,\rho^b(y)\,\rho^c(x)\,U^{de}(w')\,U^{ad}(w)\\
(15)\quad & -\,K_\beta\, f^{abc}\,\rho^d(x')\,\rho^e(x)\,\rho^h(y)\,U^{ad}(w')\,U^{be}(x)\,U^{ch}(y)\\
(16)\quad & +\,K_\beta\, f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(w')\,U^{be}(y)\,U^{ch}(x)\Big]
\end{aligned}
$$

Group I, Row II, part 2 (c.c.).

$$
\begin{aligned}
& i\mathcal{N}\int e^{-ik\cdot(w'-w)}\Big[\\
(3)\quad & +K_{\alpha'}\, f^{abc}\,\rho^h(y)\,\rho^e(x')\,\rho^d(x)\,U^{ad}(x)\,U^{be}(x')\,U^{ch}(y)\\
(4)\quad & -K_{\alpha'}\, f^{abc}\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(x)\,U^{be}(y)\,U^{ch}(x')\\
(7)\quad & -K_{\alpha'}\, f^{abc}\,\rho^h(y)\,\rho^e(x')\,\rho^d(x)\,U^{ad}(w)\,U^{be}(x')\,U^{ch}(y)\\
(8)\quad & +K_{\alpha'}\, f^{abc}\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(w)\,U^{be}(y)\,U^{ch}(x')\\
(9)\quad & +K_{\beta'}\, f^{abc}\,\rho^c(y)\,\rho^b(x')\,\rho^e(x)\,U^{de}(x)\,U^{ad}(w')\\
(10)\quad & -K_{\beta'}\, f^{abc}\,\rho^c(x')\,\rho^b(y)\,\rho^e(x)\,U^{de}(x)\,U^{ad}(w')\\
(11)\quad & -K_{\beta'}\, f^{abc}\,\rho^h(y)\,\rho^e(x')\,\rho^d(x)\,U^{ad}(x)\,U^{be}(x')\,U^{ch}(y)\\
(12)\quad & +K_{\beta'}\, f^{abc}\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(x)\,U^{be}(y)\,U^{ch}(x')\\
(13)\quad & -K_{\beta'}\, f^{abc}\,\rho^c(y)\,\rho^b(x')\,\rho^e(x)\,U^{de}(w)\,U^{ad}(w')\\
(14)\quad & +K_{\beta'}\, f^{abc}\,\rho^c(x')\,\rho^b(y)\,\rho^e(x)\,U^{de}(w)\,U^{ad}(w')\\
(15)\quad & +K_{\beta'}\, f^{abc}\,\rho^h(y)\,\rho^e(x')\,\rho^d(x)\,U^{ad}(w)\,U^{be}(x')\,U^{ch}(y)\\
(16)\quad & -K_{\beta'}\, f^{abc}\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(w)\,U^{be}(y)\,U^{ch}(x')\Big]
\end{aligned}
$$

Group I, Row III.

$$
\begin{aligned}
& i\mathcal{N}\int e^{-ik\cdot(w'-w)}\Big[\\
(1)\quad & +\,K_\alpha\, f^{abc}\,\rho^d(x')\,\rho^e(x)\,\rho^h(y)\,U^{ad}(x')\,U^{ce}(x)\,U^{bh}(z)\\
(2)\quad & -\,K_\alpha\, f^{abc}\,\rho^d(x')\,\rho^e(x)\,\rho^h(y)\,U^{ad}(x')\,U^{ce}(x)\,U^{bh}(y)\\
(3)\quad & -\,K_\alpha\, f^{abc}\,\rho^d(x')\,\rho^e(x)\,\rho^h(y)\,U^{ad}(w')\,U^{ce}(x)\,U^{bh}(z)\\
(4)\quad & +\,K_\alpha\, f^{abc}\,\rho^d(x')\,\rho^e(x)\,\rho^h(y)\,U^{ad}(w')\,U^{ce}(x)\,U^{bh}(y)\\
(5)\quad & -K_{\alpha'}\, f^{abc}\,\rho^h(y)\,\rho^e(x')\,\rho^d(x)\,U^{ad}(x)\,U^{ce}(x')\,U^{bh}(z)\\
(6)\quad & +K_{\alpha'}\, f^{abc}\,\rho^h(y)\,\rho^e(x')\,\rho^d(x)\,U^{ad}(x)\,U^{ce}(x')\,U^{bh}(y)\\
(7)\quad & +K_{\alpha'}\, f^{abc}\,\rho^h(y)\,\rho^e(x')\,\rho^d(x)\,U^{ad}(w)\,U^{ce}(x')\,U^{bh}(z)\\
(8)\quad & -K_{\alpha'}\, f^{abc}\,\rho^h(y)\,\rho^e(x')\,\rho^d(x)\,U^{ad}(w)\,U^{ce}(x')\,U^{bh}(y)\Big]
\end{aligned}
$$

$\mathcal{D}$ (replaces the γ, γ' terms of Group I, Row III).

$$
\begin{aligned}
& i\mathcal{N}\int e^{-ik\cdot(w'-w)}\Big[\\
(1)\quad & -\,2\,K_\beta\, f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(x')\,U^{ce}(y)\,U^{bh}(w)\\
(2)\quad & +\,2\,K_\beta\, f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(x')\,U^{ce}(y)\,U^{bh}(x)\\
(3)\quad & +\,2\,K_\beta\, f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(w')\,U^{ce}(y)\,U^{bh}(w)\\
(4)\quad & -\,2\,K_\beta\, f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(w')\,U^{ce}(y)\,U^{bh}(x)\\
(5)\quad & +2\,K_{\beta'}\, f^{abc}\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(x)\,U^{ce}(y)\,U^{bh}(w')\\
(6)\quad & -2\,K_{\beta'}\, f^{abc}\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(x)\,U^{ce}(y)\,U^{bh}(x')\\
(7)\quad & -2\,K_{\beta'}\, f^{abc}\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(w)\,U^{ce}(y)\,U^{bh}(w')\\
(8)\quad & +2\,K_{\beta'}\, f^{abc}\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(w)\,U^{ce}(y)\,U^{bh}(x')\Big]
\end{aligned}
$$

Group I, Row IV.

$$
\begin{aligned}
& i\mathcal{N}\int e^{-ik\cdot(w'-w)}\Big[\\
&(1)\quad +K_\alpha\, f^{abc}\,\rho^e(x')\,\rho^l(y)\,\rho^c(x)\,U^{de}(x')\,U^{ad}(w)\,U^{hb}(z)\,U^{hl}(y) \\
&(2)\quad -K_\alpha\, f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{da}(x')\,U^{be}(y)\,U^{ch}(x) \\
&(3)\quad -K_\alpha\, f^{abc}\,\rho^e(x')\,\rho^l(y)\,\rho^c(x)\,U^{de}(w')\,U^{ad}(w)\,U^{hb}(z)\,U^{hl}(y) \\
&(4)\quad +K_\alpha\, f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{da}(w')\,U^{be}(y)\,U^{ch}(x) \\
&(5)\quad -K_{\alpha'}\, f^{abc}\,\rho^c(x')\,\rho^l(y)\,\rho^e(x)\,U^{de}(x)\,U^{ad}(w')\,U^{hb}(z)\,U^{hl}(y) \\
&(6)\quad +K_{\alpha'}\, f^{abc}\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{da}(x)\,U^{ch}(x')\,U^{be}(y) \\
&(7)\quad +K_{\alpha'}\, f^{abc}\,\rho^c(x')\,\rho^l(y)\,\rho^e(x)\,U^{de}(w)\,U^{ad}(w')\,U^{hb}(z)\,U^{hl}(y) \\
&(8)\quad -K_{\alpha'}\, f^{abc}\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{da}(w)\,U^{ch}(x')\,U^{be}(y) \\
&(9)\quad -2\,K_\beta\, f^{abc}\,\rho^e(x')\,\rho^l(y)\,\rho^c(x)\,U^{de}(x')\,U^{ad}(w)\,U^{hb}(z)\,U^{hl}(y) \\
&(10)\quad +2\,K_\beta\, f^{abc}\,\rho^e(x')\,\rho^l(y)\,\rho^c(x)\,U^{de}(w')\,U^{ad}(w)\,U^{hb}(z)\,U^{hl}(y) \\
&(11)\quad +2\,K_\beta\, f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{da}(x')\,U^{be}(y)\,U^{hc}(x) \\
&(12)\quad -2\,K_\beta\, f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{da}(w')\,U^{be}(y)\,U^{hc}(x) \\
&(13)\quad +2\,K_{\beta'}\, f^{abc}\,\rho^c(x')\,\rho^l(y)\,\rho^e(x)\,U^{de}(x)\,U^{ad}(w')\,U^{hb}(z)\,U^{hl}(y) \\
&(14)\quad -2\,K_{\beta'}\, f^{abc}\,\rho^c(x')\,\rho^l(y)\,\rho^e(x)\,U^{de}(w)\,U^{ad}(w')\,U^{hb}(z)\,U^{hl}(y) \\
&(15)\quad +2\,K_{\beta'}\, f^{abc}\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{be}(y)\,U^{ad}(w)\,U^{ch}(x') \\
&(16)\quad -2\,K_{\beta'}\, f^{abc}\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{be}(y)\,U^{ad}(x)\,U^{ch}(x')\Big]
\end{aligned}
$$

Group I, Row V, part 1.

$$
\begin{aligned}
& i\mathcal{N}\int e^{-ik\cdot(w'-w)}\Big[\\
&(1)\quad -\tfrac{1}{2}\,K_\beta\, f^{abc}\,\rho^d(x')\big\{\rho^h(x),\rho^e(y)\big\}\,U^{ad}(x')\,U^{be}(z)\,U^{ch}(w) \\
&(2)\quad +\tfrac{1}{2}\,K_\beta\, f^{abc}\,\rho^d(x')\big\{\rho^h(x),\rho^e(y)\big\}\,U^{ad}(w')\,U^{be}(z)\,U^{ch}(w) \\
&(3)\quad +\tfrac{1}{2}\,K_\beta\, f^{abc}\,\rho^d(x')\big\{\rho^e(x),\rho^h(y)\big\}\,U^{ad}(x')\,U^{ce}(x)\,U^{bh}(y) \\
&(4)\quad -\tfrac{1}{2}\,K_\beta\, f^{abc}\,\rho^d(x')\big\{\rho^e(x),\rho^h(y)\big\}\,U^{ad}(w')\,U^{ce}(x)\,U^{bh}(y) \\
&(5)\quad +\tfrac{1}{2}\,K_{\beta'}\, f^{abc}\,\big\{\rho^h(x'),\rho^e(y)\big\}\rho^d(x)\,U^{ad}(x)\,U^{be}(z)\,U^{ch}(w') \\
&(6)\quad -\tfrac{1}{2}\,K_{\beta'}\, f^{abc}\,\big\{\rho^h(x'),\rho^e(y)\big\}\rho^d(x)\,U^{ad}(w)\,U^{be}(z)\,U^{ch}(w') \\
&(7)\quad +\tfrac{1}{2}\,K_{\beta'}\, f^{abc}\,\big\{\rho^e(x'),\rho^h(y)\big\}\rho^d(x)\,U^{ad}(w)\,U^{bh}(y)\,U^{ce}(x') \\
&(8)\quad -\tfrac{1}{2}\,K_{\beta'}\, f^{abc}\,\big\{\rho^e(x'),\rho^h(y)\big\}\rho^d(x)\,U^{ad}(x)\,U^{bh}(y)\,U^{ce}(x')\Big]
\end{aligned}
$$

Group I, Row V, part 2.

$$
\begin{aligned}
& i\mathcal{N}\int e^{-ik\cdot(w'-w)}\Big[\\
&(1)\quad +\tfrac{1}{2}\,K_\beta\, f^{abc}\,\rho^d(x')\big\{\rho^e(x),\rho^h(y)\big\}\,U^{ad}(x')\,U^{be}(w)\,U^{ch}(z) \\
&(2)\quad -\tfrac{1}{2}\,K_\beta\, f^{abc}\,\rho^d(x')\big\{\rho^h(x),\rho^e(y)\big\}\,U^{ad}(x')\,U^{ce}(y)\,U^{bh}(x) \\
&(3)\quad -\tfrac{1}{2}\,K_\beta\, f^{abc}\,\rho^d(x')\big\{\rho^e(x),\rho^h(y)\big\}\,U^{ad}(w')\,U^{be}(w)\,U^{ch}(z) \\
&(4)\quad +\tfrac{1}{2}\,K_\beta\, f^{abc}\,\rho^d(x')\big\{\rho^h(x),\rho^e(y)\big\}\,U^{ad}(w')\,U^{ce}(y)\,U^{bh}(x) \\
&(5)\quad -\tfrac{1}{2}\,K_{\beta'}\, f^{abc}\,\big\{\rho^e(x'),\rho^h(y)\big\}\rho^d(x)\,U^{ad}(x)\,U^{be}(w')\,U^{ch}(z) \\
&(6)\quad +\tfrac{1}{2}\,K_{\beta'}\, f^{abc}\,\big\{\rho^h(x'),\rho^e(y)\big\}\rho^d(x)\,U^{ad}(x)\,U^{ce}(y)\,U^{bh}(x') \\
&(7)\quad +\tfrac{1}{2}\,K_{\beta'}\, f^{abc}\,\big\{\rho^e(x'),\rho^h(y)\big\}\rho^d(x)\,U^{ad}(w)\,U^{ch}(z)\,U^{be}(w') \\
&(8)\quad -\tfrac{1}{2}\,K_{\beta'}\, f^{abc}\,\big\{\rho^h(x'),\rho^e(y)\big\}\rho^d(x)\,U^{ad}(w)\,U^{bh}(x')\,U^{ce}(y)\Big]
\end{aligned}
$$

Group II, Row I.

$$
\begin{aligned}
& i\mathcal{N}\int e^{-ik\cdot(w'-w)}\Big[\\
&(1)\quad -\tfrac{1}{2}\,K_\alpha\, f^{abc}\,\big\{\rho^e(x'),\rho^b(y)\big\}\rho^c(x)\,U^{de}(w')\,U^{ad}(w) \\
&(2)\quad +\tfrac{1}{2}\,K_\alpha\, f^{abc}\,\big\{\rho^d(x'),\rho^e(y)\big\}\rho^h(x)\,U^{be}(z)\,U^{ad}(w')\,U^{ch}(x) \\
&(3)\quad +\tfrac{1}{2}\,K_\alpha\, f^{abc}\,\big\{\rho^h(x'),\rho^l(y)\big\}\rho^c(x)\,U^{eh}(x')\,U^{bd}(z)\,U^{dl}(y)\,U^{ae}(w)
\end{aligned}
$$

$$
\begin{aligned}
&(4)\ \ -\tfrac{1}{2}\,K_\alpha\, f^{abc}\,\big\{\rho^d(x'),\rho^e(y)\big\}\rho^h(x)\,U^{ad}(x')\,U^{be}(y)\,U^{ch}(x)\\
&(5)\ \ +K_\beta\, f^{abc}\,\big\{\rho^e(x'),\rho^b(y)\big\}\rho^c(x)\,U^{de}(w')\,U^{ad}(w)\\
&(6)\ \ -K_\beta\, f^{abc}\,\big\{\rho^d(x'),\rho^e(y)\big\}\rho^h(x)\,U^{ad}(w')\,U^{eb}(z)\,U^{ch}(x)\\
&(7)\ \ -K_\beta\, f^{abc}\,\big\{\rho^h(x'),\rho^l(y)\big\}\rho^c(x)\,U^{eh}(x')\,U^{bd}(z)\,U^{dl}(y)\,U^{ea}(w)\\
&(8)\ \ +K_\beta\, f^{abc}\,\big\{\rho^d(x'),\rho^e(y)\big\}\rho^h(x)\,U^{da}(x')\,U^{be}(y)\,U^{ch}(x)\\
&(9)\ \ +\tfrac{1}{2}\,K_{\alpha'}\, f^{abc}\,\rho^c(x')\big\{\rho^e(x),\rho^b(y)\big\}\,U^{ad}(w')\,U^{de}(w)\\
&(10)\ \ -\tfrac{1}{2}\,K_{\alpha'}\, f^{abc}\,\rho^c(x')\big\{\rho^h(x),\rho^l(y)\big\}\,U^{ad}(w')\,U^{be}(z)\,U^{dh}(x)\,U^{el}(y)\\
&(11)\ \ -\tfrac{1}{2}\,K_{\alpha'}\, f^{abc}\,\rho^e(x')\big\{\rho^h(x),\rho^d(y)\big\}\,U^{bd}(z)\,U^{ce}(x')\,U^{ah}(w)\\
&(12)\ \ +\tfrac{1}{2}\,K_{\alpha'}\, f^{abc}\,\rho^d(x')\big\{\rho^e(x),\rho^h(y)\big\}\,U^{cd}(x')\,U^{ae}(x)\,U^{bh}(y)\\
&(13)\ \ -K_{\beta'}\, f^{abc}\,\rho^c(x')\big\{\rho^e(x),\rho^b(y)\big\}\,U^{ad}(w')\,U^{de}(w)\\
&(14)\ \ +K_{\beta'}\, f^{abc}\,\rho^c(x')\big\{\rho^h(x),\rho^l(y)\big\}\,U^{ad}(w')\,U^{be}(z)\,U^{dh}(x)\,U^{el}(y)\\
&(15)\ \ +K_{\beta'}\, f^{abc}\,\rho^e(x')\big\{\rho^h(x),\rho^d(y)\big\}\,U^{ce}(x')\,U^{db}(z)\,U^{ah}(w)\\
&(16)\ \ -K_{\beta'}\, f^{abc}\,\rho^d(x')\big\{\rho^e(x),\rho^h(y)\big\}\,U^{cd}(x')\,U^{ae}(x)\,U^{bh}(y)\Big]
\end{aligned}
$$

Group III, Row I.

$$
\begin{aligned}
&i\mathcal{N}\int e^{-ik\cdot(w'-w)}\Big[\\
&(1)\ \ -K_\alpha\, f^{abc}\,\rho^e(x')\,\rho^l(y)\,\rho^c(x)\,U^{de}(x')\,U^{hl}(y)\,U^{hb}(z)\,U^{ad}(w)\\
&(2)\ \ +K_\alpha\, f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(x')\,U^{be}(y)\,U^{ch}(x)\\
&(3)\ \ +K_\alpha\, f^{abc}\,\rho^l(x')\,\rho^e(y)\,\rho^c(x)\,U^{bd}(z)\,U^{de}(y)\,U^{hl}(w')\,U^{ah}(w)\\
&(4)\ \ -K_\alpha\, f^{abc}\,\rho^e(x')\,\rho^d(y)\,\rho^h(x)\,U^{ae}(w')\,U^{bd}(y)\,U^{ch}(x)\\
&(5)\ \ +2\,K_\beta\, f^{abc}\,\rho^e(x')\,\rho^l(y)\,\rho^c(x)\,U^{de}(x')\,U^{hl}(y)\,U^{hb}(z)\,U^{ad}(w)\\
&(6)\ \ -2\,K_\beta\, f^{abc}\,\rho^d(x')\,\rho^e(y)\,\rho^h(x)\,U^{ad}(x')\,U^{be}(y)\,U^{ch}(x)\\
&(7)\ \ -2\,K_\beta\, f^{abc}\,\rho^l(x')\,\rho^e(y)\,\rho^c(x)\,U^{bd}(z)\,U^{de}(y)\,U^{hl}(w')\,U^{ah}(w)\\
&(8)\ \ +2\,K_\beta\, f^{abc}\,\rho^e(x')\,\rho^d(y)\,\rho^h(x)\,U^{ae}(w')\,U^{bd}(y)\,U^{ch}(x)\\
&(9)\ \ +K_{\alpha'}\, f^{abc}\,\rho^c(x')\,\rho^l(y)\,\rho^e(x)\,U^{de}(x)\,U^{hl}(y)\,U^{hb}(z)\,U^{ad}(w')\\
&(10)\ \ -K_{\alpha'}\, f^{abc}\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(x)\,U^{be}(y)\,U^{ch}(x')\\
&(11)\ \ -K_{\alpha'}\, f^{abc}\,\rho^c(x')\,\rho^e(y)\,\rho^l(x)\,U^{bd}(z)\,U^{de}(y)\,U^{hl}(w)\,U^{ha}(w')\\
&(12)\ \ +K_{\alpha'}\, f^{abc}\,\rho^h(x')\,\rho^d(y)\,\rho^e(x)\,U^{bd}(y)\,U^{ae}(w)\,U^{ch}(x')\\
&(13)\ \ -2\,K_{\beta'}\, f^{abc}\,\rho^c(x')\,\rho^l(y)\,\rho^e(x)\,U^{de}(x)\,U^{hl}(y)\,U^{hb}(z)\,U^{ad}(w')\\
&(14)\ \ +2\,K_{\beta'}\, f^{abc}\,\rho^h(x')\,\rho^e(y)\,\rho^d(x)\,U^{ad}(x)\,U^{be}(y)\,U^{ch}(x')\\
&(15)\ \ +2\,K_{\beta'}\, f^{abc}\,\rho^c(x')\,\rho^e(y)\,\rho^l(x)\,U^{de}(y)\,U^{hl}(w)\,U^{db}(z)\,U^{ha}(w')\\
&(16)\ \ -2\,K_{\beta'}\, f^{abc}\,\rho^h(x')\,\rho^d(y)\,\rho^e(x)\,U^{bd}(y)\,U^{ae}(w)\,U^{ch}(x')\Big]
\end{aligned}
$$

Group III, Row II.

$$
\begin{aligned}
&i\mathcal{N}\int e^{-ik\cdot(w'-w)}\Big[\\
&(1)\ \ -K_\alpha\, f^{abc}\,\rho^e(y)\,\rho^l(x')\,\rho^c(x)\,U^{de}(y)\,U^{hl}(x')\,U^{bd}(z)\,U^{ah}(w)\\
&(2)\ \ +K_\alpha\, f^{abc}\,\rho^d(y)\,\rho^e(x')\,\rho^h(x)\,U^{bd}(y)\,U^{ae}(x')\,U^{ch}(x)\\
&(3)\ \ +K_\alpha\, f^{abc}\,\rho^b(y)\,\rho^e(x')\,\rho^c(x)\,U^{de}(x')\,U^{ad}(w)\\
&(4)\ \ -K_\alpha\, f^{abc}\,\rho^e(y)\,\rho^d(x')\,\rho^h(x)\,U^{ad}(x')\,U^{eb}(z)\,U^{ch}(x)\\
&(5)\ \ +2\,K_\beta\, f^{abc}\,\rho^e(y)\,\rho^l(x')\,\rho^c(x)\,U^{de}(y)\,U^{hl}(x')\,U^{bd}(z)\,U^{ha}(w)\\
&(6)\ \ -2\,K_\beta\, f^{abc}\,\rho^d(y)\,\rho^e(x')\,\rho^h(x)\,U^{db}(y)\,U^{ae}(x')\,U^{ch}(x)\\
&(7)\ \ -2\,K_\beta\, f^{abc}\,\rho^b(y)\,\rho^e(x')\,\rho^c(x)\,U^{de}(x')\,U^{da}(w)\\
&(8)\ \ +2\,K_\beta\, f^{abc}\,\rho^e(y)\,\rho^d(x')\,\rho^h(x)\,U^{ad}(x')\,U^{eb}(z)\,U^{ch}(x)\\
&(9)\ \ +K_{\alpha'}\, f^{abc}\,\rho^c(x')\,\rho^l(x)\,\rho^e(y)\,U^{de}(y)\,U^{hl}(x)\,U^{bd}(z)\,U^{ah}(w')\\
&(10)\ \ -K_{\alpha'}\, f^{abc}\,\rho^h(x')\,\rho^e(x)\,\rho^d(y)\,U^{bd}(y)\,U^{ae}(x)\,U^{ch}(x')\\
&(11)\ \ -K_{\alpha'}\, f^{abc}\,\rho^c(x')\,\rho^e(x)\,\rho^b(y)\,U^{de}(x)\,U^{ad}(w')\\
&(12)\ \ +K_{\alpha'}\, f^{abc}\,\rho^h(x')\,\rho^d(x)\,\rho^e(y)\,U^{ad}(x)\,U^{eb}(z)\,U^{ch}(x')\\
&(13)\ \ -2\,K_{\beta'}\, f^{abc}\,\rho^c(x')\,\rho^l(x)\,\rho^e(y)\,U^{de}(y)\,U^{hl}(x)\,U^{bd}(z)\,U^{ha}(w')
\end{aligned}
$$

$$
\begin{aligned}
(14)\quad & +2\,K_{\beta'}\,f^{abc}\,\rho^{h}(x')\,\rho^{e}(x)\,\rho^{d}(y)\,U^{bd}(y)\,U^{ae}(x)\,U^{ch}(x') \\
(15)\quad & +2\,K_{\beta'}\,f^{abc}\,\rho^{c}(x')\,\rho^{e}(x)\,\rho^{b}(y)\,U^{de}(x)\,U^{ad}(w') \\
(16)\quad & -2\,K_{\beta'}\,f^{abc}\,\rho^{h}(x')\,\rho^{d}(x)\,\rho^{e}(y)\,U^{ad}(x)\,U^{eb}(z)\,U^{ch}(x')\Big]
\end{aligned}
$$

Group III, Row III.

$$
\begin{aligned}
i\mathcal{N}\int e^{-ik\cdot(w'-w)}\Big[& \\
(1)\quad & +K_{\beta}\,f^{abc}\,\big\{\rho^{h}(x'),\rho^{e}(y)\big\}\rho^{d}(x)\,U^{ad}(w)\,U^{be}(z)\,U^{ch}(w') \\
(2)\quad & -K_{\beta}\,f^{abc}\,\big\{\rho^{e}(x'),\rho^{h}(y)\big\}\rho^{d}(x)\,U^{ad}(w)\,U^{ce}(x')\,U^{bh}(y) \\
(3)\quad & -K_{\beta}\,f^{abc}\,\big\{\rho^{h}(x'),\rho^{d}(y)\big\}\rho^{e}(x)\,U^{db}(z)\,U^{ae}(x)\,U^{ch}(w') \\
(4)\quad & +K_{\beta}\,f^{abc}\,\big\{\rho^{e}(x'),\rho^{h}(y)\big\}\rho^{d}(x)\,U^{ad}(x)\,U^{ce}(x')\,U^{bh}(y) \\
(5)\quad & -K_{\beta'}\,f^{abc}\,\rho^{d}(x')\big\{\rho^{h}(x),\rho^{e}(y)\big\}\,U^{ad}(w')\,U^{be}(z)\,U^{ch}(w) \\
(6)\quad & +K_{\beta'}\,f^{abc}\,\rho^{d}(x')\big\{\rho^{e}(x),\rho^{h}(y)\big\}\,U^{ad}(w')\,U^{ce}(x)\,U^{bh}(y) \\
(7)\quad & +K_{\beta'}\,f^{abc}\,\rho^{e}(x')\big\{\rho^{h}(x),\rho^{d}(y)\big\}\,U^{bd}(z)\,U^{ae}(x')\,U^{ch}(w) \\
(8)\quad & -K_{\beta'}\,f^{abc}\,\rho^{d}(x')\big\{\rho^{e}(x),\rho^{h}(y)\big\}\,U^{ad}(x')\,U^{ce}(x)\,U^{bh}(y)\Big]
\end{aligned}
$$

Group III, Row V.

$$
\begin{aligned}
i\mathcal{N}\int e^{-ik\cdot(w'-w)}\Big[& \\
(1)\quad & +2\,K_{\beta}\,f^{abc}\,\rho^{h}(x')\,\rho^{e}(y)\,\rho^{d}(x)\,U^{ad}(w)\,U^{be}(y)\,U^{ch}(x') \\
(2)\quad & -2\,K_{\beta}\,f^{abc}\,\rho^{e}(x')\,\rho^{h}(y)\,\rho^{d}(x)\,U^{ad}(w)\,U^{ce}(w')\,U^{bh}(y) \\
(3)\quad & -2\,K_{\beta}\,f^{abc}\,\rho^{h}(x')\,\rho^{e}(y)\,\rho^{d}(x)\,U^{be}(y)\,U^{ad}(x)\,U^{ch}(x') \\
(4)\quad & +2\,K_{\beta}\,f^{abc}\,\rho^{e}(x')\,\rho^{h}(y)\,\rho^{d}(x)\,U^{ad}(x)\,U^{ce}(w')\,U^{bh}(y) \\
(5)\quad & -2\,K_{\beta'}\,f^{abc}\,\rho^{d}(x')\,\rho^{e}(y)\,\rho^{h}(x)\,U^{ad}(w')\,U^{be}(y)\,U^{ch}(x) \\
(6)\quad & +2\,K_{\beta'}\,f^{abc}\,\rho^{d}(x')\,\rho^{h}(y)\,\rho^{e}(x)\,U^{ad}(w')\,U^{ce}(w)\,U^{bh}(y) \\
(7)\quad & +2\,K_{\beta'}\,f^{abc}\,\rho^{d}(x')\,\rho^{e}(y)\,\rho^{h}(x)\,U^{be}(y)\,U^{ad}(x')\,U^{ch}(x) \\
(8)\quad & -2\,K_{\beta'}\,f^{abc}\,\rho^{d}(x')\,\rho^{h}(y)\,\rho^{e}(x)\,U^{ad}(x')\,U^{ce}(w)\,U^{bh}(y)\Big]
\end{aligned}
$$

Group III, Row VI.

$$
\begin{aligned}
i\mathcal{N}\int e^{-ik\cdot(w'-w)}\Big[& \\
(1)\quad & +2\,K_{\beta}\,f^{abc}\,\rho^{h}(y)\,\rho^{e}(x')\,\rho^{d}(x)\,U^{ad}(w)\,U^{ce}(x')\,U^{bh}(y) \\
(2)\quad & -2\,K_{\beta}\,f^{abc}\,\rho^{e}(y)\,\rho^{h}(x')\,\rho^{d}(x)\,U^{ad}(w)\,U^{be}(z)\,U^{ch}(x') \\
(3)\quad & -2\,K_{\beta}\,f^{abc}\,\rho^{h}(y)\,\rho^{e}(x')\,\rho^{d}(x)\,U^{bh}(y)\,U^{ad}(x)\,U^{ce}(x') \\
(4)\quad & +2\,K_{\beta}\,f^{abc}\,\rho^{d}(y)\,\rho^{h}(x')\,\rho^{e}(x)\,U^{ae}(x)\,U^{ch}(x')\,U^{db}(z) \\
(5)\quad & -2\,K_{\beta'}\,f^{abc}\,\rho^{d}(x')\,\rho^{e}(x)\,\rho^{h}(y)\,U^{ad}(w')\,U^{bh}(y)\,U^{ce}(x) \\
(6)\quad & +2\,K_{\beta'}\,f^{abc}\,\rho^{d}(x')\,\rho^{h}(x)\,\rho^{e}(y)\,U^{ad}(w')\,U^{be}(z)\,U^{ch}(x) \\
(7)\quad & +2\,K_{\beta'}\,f^{abc}\,\rho^{d}(x')\,\rho^{e}(x)\,\rho^{h}(y)\,U^{bh}(y)\,U^{ad}(x')\,U^{ce}(x) \\
(8)\quad & -2\,K_{\beta'}\,f^{abc}\,\rho^{e}(x')\,\rho^{h}(x)\,\rho^{d}(y)\,U^{ae}(x')\,U^{ch}(x)\,U^{bd}(z)\Big]
\end{aligned}
$$

7 The two-ρ terms

Every term of Sec. 6.3 is reordered with Eqs. (2.1)–(2.3). The colour algebra is done, and equal terms within each source are collected. The result is a list of entries.

S: the kernel structure S_n (Table 3);

coeff.: the coefficient in units of $\mathcal{N}$;

colour structure: the colour structure, with the two charges as a symmetric (commuting) product.

Each entry is $\mathcal{N}\int e^{-ik\cdot(w'-w)}\int_{x,x',z}$(coeff.) × (kernel) × (colour structure). The c.c. is included. These are the same as the corresponding entries of `TwoRho_all.pdf`, Sec. 5.

S_1	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$
S_2	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$
S_3	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$
S_4	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(w,w,z)$
S_5	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(w',w',z)$
S_6	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(w,w',z)$
S_7	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$
S_8	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$
S_9	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$
S_{10}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w',z)$
S_{11}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$
S_{12}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$
S_{13}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$

Table 3: The kernel structures of the two-ρ terms.

7.1 From Group I, Row II

S	kernel	coeff.	colour structure
S_3	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$-2N_c$	$U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$
S_3	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$+1$	$f^{abc}\,f^{deh}\,U^{eb}(w')\,U^{lc}(x)\,U^{ad}(x')\,\rho^h(x')\,\rho^l(x)$
S_3	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$+1$	$f^{abc}\,f^{deh}\,U^{lc}(x')\,U^{eb}(w)\,U^{ad}(x)\,\rho^l(x')\,\rho^h(x)$
S_8	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	-1	$f^{abc}\,f^{ade}\,U^{hb}(w')\,U^{hd}(x)\,\rho^c(x')\,\rho^e(x)$
S_8	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	$+N_c$	$U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$
S_8	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	-1	$f^{abc}\,f^{dce}\,U^{hd}(w)\,U^{ah}(w')\,\rho^b(x')\,\rho^e(x)$
S_8	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	$+1$	$f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{bl}(x')\,U^{ce}(x)\,\rho^l(x')\,\rho^h(x)$
S_9	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$	-1	$f^{abc}\,f^{ade}\,U^{hb}(x')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)$
S_9	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$	$+N_c$	$U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')$
S_9	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$	$+1$	$f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ah}(w)\,\rho^b(x)\,\rho^e(x')$
S_9	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$	-1	$f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bl}(x)\,U^{cd}(x')\,\rho^l(x)\,\rho^h(x')$
S_{11}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$+1$	$f^{abc}\,f^{deh}\,U^{eb}(w')\,U^{dc}(x')\,U^{al}(x)\,\rho^l(x)\,\rho^h(x')$
S_{11}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	-1	$f^{abc}\,f^{deh}\,U^{ec}(x')\,U^{db}(w)\,U^{al}(x)\,\rho^h(x')\,\rho^l(x)$
S_{11}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$+2N_c$	$U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$
S_{12}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	-1	$f^{abc}\,f^{bde}\,U^{hd}(x')\,U^{ah}(w)\,\rho^e(x')\,\rho^c(x)$
S_{12}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	$-N_c$	$U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$
S_{12}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	$+1$	$f^{abc}\,f^{bde}\,U^{hd}(w')\,U^{ah}(w)\,\rho^e(x')\,\rho^c(x)$
S_{12}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	-1	$f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bd}(x')\,U^{cl}(x)\,\rho^h(x')\,\rho^l(x)$
S_{13}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	$+1$	$f^{abc}\,f^{dbe}\,U^{hd}(x')\,U^{ah}(w')\,\rho^c(x)\,\rho^e(x')$
S_{13}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	$-N_c$	$U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')$
S_{13}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	-1	$f^{abc}\,f^{dbe}\,U^{hd}(w)\,U^{ah}(w')\,\rho^c(x)\,\rho^e(x')$
S_{13}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	$+1$	$f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{be}(x')\,U^{cl}(x)\,\rho^l(x)\,\rho^h(x')$

7.2 From Group I, Row III (part 1)

S	kernel	coeff.	colour structure
S_1	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$	$+\frac{N_c}{2}$	$U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$
S_1	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$	$+\frac{1}{2}$	$f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(w')\,U^{hb}(x)\,\rho^l(x')\,\rho^c(x)$
S_1	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$	$-\frac{N_c}{2}$	$U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$
S_1	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$	$-\frac{1}{2}$	$f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(x')\,U^{hb}(x)\,\rho^l(x')\,\rho^c(x)$
S_2	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$	$+\frac{N_c}{2}$	$U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$
S_2	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$	$+\frac{1}{2}$	$f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(w)\,U^{hb}(x')\,\rho^c(x')\,\rho^l(x)$
S_2	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$	$-\frac{N_c}{2}$	$U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$
S_2	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$	$-\frac{1}{2}$	$f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(x)\,U^{hb}(x')\,\rho^c(x')\,\rho^l(x)$
S_3	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$+N_c$	$U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$
S_3	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$-\frac{1}{2}$	$f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(x)\,U^{hl}(x')\,\rho^l(x')\,\rho^c(x)$
S_3	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$-\frac{1}{2}$	$f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(x')\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)$
S_3	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$-\frac{1}{2}$	$f^{abc}\,f^{deh}\,U^{eb}(w')\,U^{lc}(x)\,U^{ad}(x')\,\rho^h(x')\,\rho^l(x)$
S_3	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$-\frac{1}{2}$	$f^{abc}\,f^{deh}\,U^{lc}(x')\,U^{eb}(w)\,U^{ad}(x)\,\rho^l(x')\,\rho^h(x)$
S_3	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$-\frac{1}{2}$	$f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x')$
S_3	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$+\frac{1}{2}$	$f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x)$
S_{11}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$-\frac{1}{2}$	$f^{abc}\,f^{deh}\,U^{eb}(w')\,U^{dc}(x')\,U^{al}(x)\,\rho^l(x)\,\rho^h(x')$
S_{11}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$+\frac{1}{2}$	$f^{abc}\,f^{deh}\,U^{ec}(x')\,U^{db}(w)\,U^{al}(x)\,\rho^h(x')\,\rho^l(x)$
S_{11}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$-N_c$	$U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$
S_{11}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$+N_c$	$U^{ab}(z)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$

S	kernel	coeff.	colour structure
S_{11}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$-\frac{1}{2}$	$f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(z)\,\rho^{h}(x')\,\rho^{l}(x)$
S_{11}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$+\frac{1}{2}$	$f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x')\,U^{bl}(z)\,\rho^{l}(x)\,\rho^{h}(x')$

7.3 From $\mathcal{D}$

S	kernel	coeff.	colour structure
S_{7}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	$+N_c$	$U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)$
S_{7}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	$-N_c$	$U^{ab}(w')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)$
S_{7}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	-1	$f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(x)\,\rho^{l}(x')\,\rho^{h}(x)$
S_{7}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	$+1$	$f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(w')\,U^{be}(x)\,\rho^{l}(x')\,\rho^{h}(x)$
S_{8}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	$-N_c$	$U^{ab}(w')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)$
S_{8}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	$+N_c$	$U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)$
S_{8}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	$+1$	$f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{bl}(x')\,U^{ce}(x)\,\rho^{l}(x')\,\rho^{h}(x)$
S_{8}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	-1	$f^{abc}\,f^{deh}\,U^{al}(w')\,U^{cd}(w)\,U^{be}(x)\,\rho^{l}(x')\,\rho^{h}(x)$
S_{9}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$	$-N_c$	$U^{ab}(x')\,U^{ac}(w)\,\rho^{b}(x')\,\rho^{c}(x)$
S_{9}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$	$+N_c$	$U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')$
S_{9}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$	-1	$f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bl}(x)\,U^{cd}(x')\,\rho^{l}(x)\,\rho^{h}(x')$
S_{9}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$	$+1$	$f^{abc}\,f^{deh}\,U^{al}(w)\,U^{ce}(w')\,U^{bd}(x')\,\rho^{h}(x')\,\rho^{l}(x)$
S_{10}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w',z)$	$+N_c$	$U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)$
S_{10}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w',z)$	$-N_c$	$U^{ab}(w)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)$
S_{10}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w',z)$	$+1$	$f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(x')\,\rho^{l}(x)\,\rho^{h}(x')$
S_{10}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w',z)$	-1	$f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(w)\,U^{bd}(x')\,\rho^{h}(x')\,\rho^{l}(x)$
S_{12}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	$-N_c$	$U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)$
S_{12}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	-1	$f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bd}(x')\,U^{cl}(x)\,\rho^{h}(x')\,\rho^{l}(x)$
S_{12}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	-1	$f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x')\,U^{bl}(x)\,\rho^{l}(x)\,\rho^{h}(x')$
S_{12}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	$+1$	$f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(w')\,U^{bl}(x)\,\rho^{l}(x)\,\rho^{h}(x')$
S_{13}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	$-N_c$	$U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')$
S_{13}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	$+1$	$f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{be}(x')\,U^{cl}(x)\,\rho^{l}(x)\,\rho^{h}(x')$
S_{13}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	$+1$	$f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(x)\,\rho^{h}(x')\,\rho^{l}(x)$
S_{13}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	-1	$f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(w)\,U^{bl}(x)\,\rho^{l}(x)\,\rho^{h}(x')$

7.4 From Group I, Row IV

S	kernel	coeff.	colour structure
S_{1}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$	$-\frac{N_c}{2}$	$U^{ab}(w')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)$
S_{1}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$	$+\frac{N_c}{2}$	$U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)$
S_{1}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$	$+\frac{1}{2}$	$f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ah}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^{l}(x')\,\rho^{e}(x)$
S_{1}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$	$-\frac{1}{2}$	$f^{abc}\,f^{cde}\,U^{hl}(w')\,U^{ah}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^{l}(x')\,\rho^{e}(x)$
S_{2}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$	$-\frac{N_c}{2}$	$U^{ab}(x')\,U^{ac}(w)\,\rho^{b}(x')\,\rho^{c}(x)$
S_{2}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$	$+\frac{N_c}{2}$	$U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)$
S_{2}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$	$-\frac{1}{2}$	$f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ah}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^{e}(x')\,\rho^{l}(x)$
S_{2}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$	$+\frac{1}{2}$	$f^{abc}\,f^{dce}\,U^{hl}(w)\,U^{ah}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^{e}(x')\,\rho^{l}(x)$
S_{3}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$+N_c$	$U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)$
S_{3}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$-\frac{1}{2}$	$f^{abc}\,f^{deh}\,U^{eb}(w')\,U^{lc}(x)\,U^{ad}(x')\,\rho^{h}(x')\,\rho^{l}(x)$
S_{3}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$-\frac{1}{2}$	$f^{abc}\,f^{deh}\,U^{lc}(x')\,U^{eb}(w)\,U^{ad}(x)\,\rho^{l}(x')\,\rho^{h}(x)$
S_{3}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$+\frac{1}{2}$	$f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^{l}(x)\,\rho^{h}(x')$
S_{3}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$-\frac{1}{2}$	$f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^{l}(x')\,\rho^{h}(x)$
S_{3}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$-\frac{1}{2}$	$f^{abc}\,f^{deh}\,U^{le}(w')\,U^{al}(w)\,U^{mb}(z)\,U^{md}(x')\,\rho^{h}(x')\,\rho^{c}(x)$
S_{3}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$+\frac{1}{2}$	$f^{abc}\,f^{deh}\,U^{ld}(w)\,U^{al}(w')\,U^{mb}(z)\,U^{me}(x)\,\rho^{c}(x')\,\rho^{h}(x)$
S_{7}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	$-N_c$	$U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)$
S_{7}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	$+N_c$	$U^{ab}(w')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)$
S_{7}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	-1	$f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ah}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^{l}(x')\,\rho^{e}(x)$
S_{7}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	$+1$	$f^{abc}\,f^{cde}\,U^{hl}(w')\,U^{ah}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^{l}(x')\,\rho^{e}(x)$
S_{8}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	$+1$	$f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(w')\,U^{hl}(x)\,\rho^{c}(x')\,\rho^{l}(x)$
S_{8}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	$-N_c$	$U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)$
S_{8}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	-1	$f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{bl}(x')\,U^{ce}(x)\,\rho^{l}(x')\,\rho^{h}(x)$
S_{8}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	-1	$f^{abc}\,f^{deh}\,U^{ld}(w)\,U^{al}(w')\,U^{mb}(z)\,U^{me}(x)\,\rho^{c}(x')\,\rho^{h}(x)$
S_{9}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$	$+1$	$f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(w)\,U^{hl}(x')\,\rho^{l}(x')\,\rho^{c}(x)$
S_{9}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$	$-N_c$	$U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')$
S_{9}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$	$+1$	$f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bl}(x)\,U^{cd}(x')\,\rho^{l}(x)\,\rho^{h}(x')$
S_{9}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$	$+1$	$f^{abc}\,f^{deh}\,U^{le}(w')\,U^{al}(w)\,U^{mb}(z)\,U^{md}(x')\,\rho^{h}(x')\,\rho^{c}(x)$
S_{10}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w',z)$	$-N_c$	$U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)$

S	kernel	coeff.	colour structure
S_{10}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w',z)$	$+N_c$	$U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$
S_{10}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w',z)$	$+1$	$f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ah}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x)$
S_{10}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w',z)$	-1	$f^{abc}\,f^{dce}\,U^{hl}(w)\,U^{ah}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x)$
S_{11}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$-\frac{1}{2}$	$f^{abc}\,f^{deh}\,U^{eb}(w')\,U^{dc}(x')\,U^{al}(x)\,\rho^l(x)\,\rho^h(x')$
S_{11}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$+\frac{1}{2}$	$f^{abc}\,f^{deh}\,U^{ec}(x')\,U^{db}(w)\,U^{al}(x)\,\rho^h(x')\,\rho^l(x)$
S_{11}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$-N_c$	$U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$
S_{11}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$+\frac{1}{2}$	$f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ah}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')$
S_{11}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$-\frac{1}{2}$	$f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ah}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')$
S_{11}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$-\frac{1}{2}$	$f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ah}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')$
S_{11}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$+\frac{1}{2}$	$f^{abc}\,f^{dce}\,U^{hd}(w)\,U^{ah}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')$
S_{12}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	$+N_c$	$U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$
S_{12}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	$+1$	$f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bd}(x')\,U^{cl}(x)\,\rho^h(x')\,\rho^l(x)$
S_{12}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	-1	$f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ah}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')$
S_{12}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	$+1$	$f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ah}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')$
S_{13}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	$+N_c$	$U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')$
S_{13}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	-1	$f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{be}(x')\,U^{cl}(x)\,\rho^l(x)\,\rho^h(x')$
S_{13}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	$+1$	$f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ah}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')$
S_{13}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	-1	$f^{abc}\,f^{dce}\,U^{hd}(w)\,U^{ah}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')$

7.5 From Group I, Row V

S	kernel	coeff.	colour structure
S_8	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	$+1$	$f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(x)\,U^{hl}(w')\,\rho^l(x')\,\rho^c(x)$
S_8	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	$-N_c$	$U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$
S_8	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	-1	$f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{bl}(x')\,U^{ce}(x)\,\rho^l(x')\,\rho^h(x)$
S_8	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	-1	$f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{be}(z)\,U^{cl}(w')\,\rho^l(x')\,\rho^h(x)$
S_9	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$	$+1$	$f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(x')\,U^{hl}(w)\,\rho^c(x')\,\rho^l(x)$
S_9	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$	$-N_c$	$U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')$
S_9	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$	$+1$	$f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bl}(x)\,U^{cd}(x')\,\rho^l(x)\,\rho^h(x')$
S_9	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$	$+1$	$f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bd}(z)\,U^{cl}(w)\,\rho^l(x)\,\rho^h(x')$
S_{12}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	$+N_c$	$U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$
S_{12}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	$+1$	$f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bd}(x')\,U^{cl}(x)\,\rho^h(x')\,\rho^l(x)$
S_{12}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	-1	$f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{bl}(z)\,U^{cd}(w)\,\rho^h(x')\,\rho^l(x)$
S_{12}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	$+1$	$f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bl}(z)\,U^{cd}(w)\,\rho^h(x')\,\rho^l(x)$
S_{13}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	$+N_c$	$U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')$
S_{13}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	-1	$f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{be}(x')\,U^{cl}(x)\,\rho^l(x)\,\rho^h(x')$
S_{13}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	$+1$	$f^{abc}\,f^{deh}\,U^{ad}(x')\,U^{bl}(z)\,U^{ce}(w')\,\rho^l(x)\,\rho^h(x')$
S_{13}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	-1	$f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{bl}(z)\,U^{ce}(w')\,\rho^l(x)\,\rho^h(x')$

7.6 From Group II, Row I

S	kernel	coeff.	colour structure
S_1	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$	$+\frac{1}{2}$	$f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(w')\,U^{hb}(x)\,\rho^l(x')\,\rho^c(x)$
S_1	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$	$+\frac{N_c}{2}$	$U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$
S_1	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$	$+\frac{N_c}{2}$	$U^{ab}(w')\,U^{bc}(w)\,\rho^a(x')\,\rho^c(x)$
S_1	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$	$+\frac{1}{2}$	$f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ah}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x)$
S_2	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$	$+\frac{1}{2}$	$f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(w)\,U^{hb}(x')\,\rho^c(x')\,\rho^l(x)$
S_2	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$	$+\frac{N_c}{2}$	$U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$
S_2	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$	$+\frac{N_c}{2}$	$U^{ab}(w')\,U^{cb}(w)\,\rho^a(x')\,\rho^c(x)$
S_2	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$	$-\frac{1}{2}$	$f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ah}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x)$
S_7	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	$-N_c$	$U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$
S_7	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	$-N_c$	$U^{ab}(w')\,U^{ca}(w)\,\rho^b(x')\,\rho^c(x)$
S_7	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	-1	$f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ah}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x)$
S_7	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	-1	$f^{abc}\,f^{deh}\,U^{al}(w')\,U^{eb}(z)\,U^{cd}(x)\,\rho^l(x')\,\rho^h(x)$
S_{10}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w',z)$	$-N_c$	$U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$
S_{10}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w',z)$	$-N_c$	$U^{ab}(w)\,U^{ca}(w')\,\rho^c(x')\,\rho^b(x)$
S_{10}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w',z)$	$+1$	$f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ah}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x)$
S_{10}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w',z)$	$+1$	$f^{abc}\,f^{deh}\,U^{ce}(x')\,U^{db}(z)\,U^{al}(w)\,\rho^l(x)\,\rho^h(x')$
S_{11}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$+\frac{1}{2}$	$f^{abc}\,f^{cde}\,U^{dh}(w')\,U^{hb}(w)\,\rho^a(x)\,\rho^e(x')$
S_{11}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$-\frac{1}{2}$	$f^{abc}\,f^{dce}\,U^{bh}(w')\,U^{dh}(w)\,\rho^a(x)\,\rho^e(x')$
S_{11}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$-N_c$	$U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$
S_{11}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$+\frac{1}{2}$	$f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(z)\,\rho^h(x')\,\rho^l(x)$

S	kernel	coeff.	colour structure
S_{11}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$-\frac{1}{2}$	$f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x')\,U^{bl}(z)\,\rho^{l}(x)\,\rho^{h}(x')$
S_{11}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$+\frac{1}{2}$	$f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ah}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')$
S_{11}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$-\frac{1}{2}$	$f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ah}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')$
S_{12}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	$+N_c$	$U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)$
S_{12}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	-1	$f^{abc}\,f^{bde}\,U^{hd}(w')\,U^{ah}(w)\,\rho^{e}(x')\,\rho^{c}(x)$
S_{12}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	-1	$f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ah}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')$
S_{12}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	-1	$f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{lb}(z)\,U^{cd}(x')\,\rho^{l}(x)\,\rho^{h}(x')$
S_{13}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	$+N_c$	$U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')$
S_{13}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	$+1$	$f^{abc}\,f^{dbe}\,U^{hd}(w)\,U^{ah}(w')\,\rho^{c}(x)\,\rho^{e}(x')$
S_{13}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	$+1$	$f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ah}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')$
S_{13}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	$+1$	$f^{abc}\,f^{deh}\,U^{ce}(x')\,U^{lb}(z)\,U^{ad}(w)\,\rho^{h}(x')\,\rho^{l}(x)$

7.7 From Group III, Row I

S	kernel	coeff.	colour structure
S_1	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$	$+\frac{N_c}{2}$	$U^{ab}(w')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)$
S_1	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$	$-\frac{N_c}{2}$	$U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)$
S_1	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$	$-\frac{1}{2}$	$f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ah}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^{l}(x')\,\rho^{e}(x)$
S_1	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$	$+\frac{1}{2}$	$f^{abc}\,f^{cde}\,U^{hl}(w')\,U^{ah}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^{l}(x')\,\rho^{e}(x)$
S_2	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$	$+\frac{N_c}{2}$	$U^{ab}(x')\,U^{ac}(w)\,\rho^{b}(x')\,\rho^{c}(x)$
S_2	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$	$-\frac{N_c}{2}$	$U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)$
S_2	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$	$+\frac{1}{2}$	$f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ah}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^{e}(x')\,\rho^{l}(x)$
S_2	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$	$-\frac{1}{2}$	$f^{abc}\,f^{dce}\,U^{hl}(w)\,U^{ah}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^{e}(x')\,\rho^{l}(x)$
S_3	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$-N_c$	$U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)$
S_3	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$+\frac{1}{2}$	$f^{abc}\,f^{deh}\,U^{eb}(w')\,U^{lc}(x)\,U^{ad}(x')\,\rho^{h}(x')\,\rho^{l}(x)$
S_3	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$+\frac{1}{2}$	$f^{abc}\,f^{deh}\,U^{lc}(x')\,U^{eb}(w)\,U^{ad}(x)\,\rho^{l}(x')\,\rho^{h}(x)$
S_3	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$-\frac{1}{2}$	$f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^{l}(x)\,\rho^{h}(x')$
S_3	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$+\frac{1}{2}$	$f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^{l}(x')\,\rho^{h}(x)$
S_3	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$+\frac{1}{2}$	$f^{abc}\,f^{deh}\,U^{le}(w')\,U^{al}(w)\,U^{mb}(z)\,U^{md}(x')\,\rho^{h}(x')\,\rho^{c}(x)$
S_3	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$-\frac{1}{2}$	$f^{abc}\,f^{deh}\,U^{ld}(w)\,U^{al}(w')\,U^{mb}(z)\,U^{me}(x)\,\rho^{c}(x')\,\rho^{h}(x)$
S_7	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	$+N_c$	$U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)$
S_7	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	$-N_c$	$U^{ab}(w')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)$
S_7	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	$+1$	$f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ah}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^{l}(x')\,\rho^{e}(x)$
S_7	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	-1	$f^{abc}\,f^{cde}\,U^{hl}(w')\,U^{ah}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^{l}(x')\,\rho^{e}(x)$
S_8	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	-1	$f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(w')\,U^{hl}(x)\,\rho^{c}(x')\,\rho^{l}(x)$
S_8	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	$+N_c$	$U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)$
S_8	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	$+1$	$f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{bl}(x')\,U^{ce}(x)\,\rho^{l}(x')\,\rho^{h}(x)$
S_8	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	$+1$	$f^{abc}\,f^{deh}\,U^{ld}(w)\,U^{al}(w')\,U^{mb}(z)\,U^{me}(x)\,\rho^{c}(x')\,\rho^{h}(x)$
S_9	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$	-1	$f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(w)\,U^{hl}(x')\,\rho^{l}(x')\,\rho^{c}(x)$
S_9	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$	$+N_c$	$U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')$
S_9	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$	-1	$f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bl}(x)\,U^{cd}(x')\,\rho^{l}(x)\,\rho^{h}(x')$
S_9	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$	-1	$f^{abc}\,f^{deh}\,U^{le}(w')\,U^{al}(w)\,U^{mb}(z)\,U^{md}(x')\,\rho^{h}(x')\,\rho^{c}(x)$
S_{10}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w',z)$	$+N_c$	$U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)$
S_{10}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w',z)$	$-N_c$	$U^{ab}(w)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)$
S_{10}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w',z)$	-1	$f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ah}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^{e}(x')\,\rho^{l}(x)$
S_{10}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w',z)$	$+1$	$f^{abc}\,f^{dce}\,U^{hl}(w)\,U^{ah}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^{e}(x')\,\rho^{l}(x)$
S_{11}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$+\frac{1}{2}$	$f^{abc}\,f^{deh}\,U^{eb}(w')\,U^{dc}(x')\,U^{al}(x)\,\rho^{l}(x)\,\rho^{h}(x')$
S_{11}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$-\frac{1}{2}$	$f^{abc}\,f^{deh}\,U^{ec}(x')\,U^{db}(w)\,U^{al}(x)\,\rho^{h}(x')\,\rho^{l}(x)$
S_{11}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$+N_c$	$U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)$
S_{11}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$-\frac{1}{2}$	$f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ah}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')$
S_{11}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$+\frac{1}{2}$	$f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ah}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')$
S_{11}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$+\frac{1}{2}$	$f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ah}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')$
S_{11}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$-\frac{1}{2}$	$f^{abc}\,f^{dce}\,U^{hd}(w)\,U^{ah}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')$
S_{12}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	$-N_c$	$U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)$
S_{12}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	-1	$f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bd}(x')\,U^{cl}(x)\,\rho^{h}(x')\,\rho^{l}(x)$
S_{12}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	$+1$	$f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ah}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')$
S_{12}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	-1	$f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ah}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')$
S_{13}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	$-N_c$	$U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')$
S_{13}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	$+1$	$f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{be}(x')\,U^{cl}(x)\,\rho^{l}(x)\,\rho^{h}(x')$
S_{13}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	-1	$f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ah}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')$
S_{13}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	$+1$	$f^{abc}\,f^{dce}\,U^{hd}(w)\,U^{ah}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')$

7.8 From Group III, Row II

S	kernel	coeff.	colour structure
S_1	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$	$-\frac{N_c}{2}$	$U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$
S_1	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$	$-\frac{1}{2}$	$f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(x')\,U^{hb}(x)\,\rho^l(x')\,\rho^c(x)$
S_1	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$	$-\frac{N_c}{2}$	$U^{ab}(w)\,U^{ca}(x')\,\rho^c(x')\,\rho^b(x)$
S_1	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$	$-\frac{1}{2}$	$f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ah}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x)$
S_2	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$	$-\frac{N_c}{2}$	$U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$
S_2	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$	$-\frac{1}{2}$	$f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(x)\,U^{hb}(x')\,\rho^c(x')\,\rho^l(x)$
S_2	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$	$-\frac{N_c}{2}$	$U^{ab}(w')\,U^{ca}(x)\,\rho^b(x')\,\rho^c(x)$
S_2	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$	$+\frac{1}{2}$	$f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ah}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x)$
S_3	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$+N_c$	$U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$
S_3	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$-\frac{1}{2}$	$f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(x)\,U^{hl}(x')\,\rho^l(x')\,\rho^c(x)$
S_3	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$-\frac{1}{2}$	$f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(x')\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)$
S_3	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$+\frac{1}{2}$	$f^{abc}\,f^{ade}\,U^{hb}(w)\,U^{dh}(x')\,\rho^e(x')\,\rho^c(x)$
S_3	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$-\frac{1}{2}$	$f^{abc}\,f^{dae}\,U^{hb}(w')\,U^{dh}(x)\,\rho^c(x')\,\rho^e(x)$
S_3	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$+\frac{1}{2}$	$f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x')$
S_3	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$-\frac{1}{2}$	$f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x)$
S_7	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	$+N_c$	$U^{ab}(x')\,U^{ca}(w)\,\rho^b(x')\,\rho^c(x)$
S_7	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	$+N_c$	$U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$
S_7	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	$+1$	$f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ah}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x)$
S_7	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	$+1$	$f^{abc}\,f^{deh}\,U^{al}(x')\,U^{eb}(z)\,U^{cd}(x)\,\rho^l(x')\,\rho^h(x)$
S_8	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	-1	$f^{abc}\,f^{ade}\,U^{hb}(w')\,U^{hd}(x)\,\rho^c(x')\,\rho^e(x)$
S_8	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	$+1$	$f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(w')\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)$
S_8	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	$-N_c$	$U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$
S_8	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	-1	$f^{abc}\,f^{deh}\,U^{ae}(x)\,U^{db}(z)\,U^{cl}(x')\,\rho^l(x')\,\rho^h(x)$
S_9	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$	-1	$f^{abc}\,f^{ade}\,U^{hb}(x')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)$
S_9	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$	$+1$	$f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(w)\,U^{hl}(x')\,\rho^l(x')\,\rho^c(x)$
S_9	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$	$-N_c$	$U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')$
S_9	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$	$+1$	$f^{abc}\,f^{deh}\,U^{ad}(x')\,U^{eb}(z)\,U^{cl}(x)\,\rho^h(x')\,\rho^l(x)$
S_{10}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w',z)$	$+N_c$	$U^{ab}(x)\,U^{ca}(w')\,\rho^c(x')\,\rho^b(x)$
S_{10}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w',z)$	$+N_c$	$U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$
S_{10}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w',z)$	-1	$f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ah}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x)$
S_{10}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w',z)$	-1	$f^{abc}\,f^{deh}\,U^{al}(x)\,U^{db}(z)\,U^{ce}(x')\,\rho^l(x)\,\rho^h(x')$
S_{11}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$-\frac{1}{2}$	$f^{abc}\,f^{cde}\,U^{hb}(w)\,U^{dh}(x')\,\rho^a(x)\,\rho^e(x')$
S_{11}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$+\frac{1}{2}$	$f^{abc}\,f^{dce}\,U^{hb}(w')\,U^{dh}(x')\,\rho^a(x)\,\rho^e(x')$
S_{11}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$+N_c$	$U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$
S_{11}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$-N_c$	$U^{ab}(z)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$
S_{11}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$-\frac{1}{2}$	$f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ah}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')$
S_{11}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$+\frac{1}{2}$	$f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ah}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')$
S_{12}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	$+1$	$f^{abc}\,f^{bde}\,U^{hd}(x')\,U^{ah}(w)\,\rho^e(x')\,\rho^c(x)$
S_{12}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	$-N_c$	$U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$
S_{12}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	$+1$	$f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ah}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')$
S_{12}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	$+N_c$	$U^{ab}(z)\,U^{bc}(x')\,\rho^a(x)\,\rho^c(x')$
S_{13}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	-1	$f^{abc}\,f^{dbe}\,U^{hd}(x')\,U^{ah}(w')\,\rho^c(x)\,\rho^e(x')$
S_{13}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	$-N_c$	$U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')$
S_{13}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	-1	$f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ah}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')$
S_{13}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	$+N_c$	$U^{ab}(z)\,U^{bc}(x')\,\rho^c(x')\,\rho^a(x)$

7.9 From Group III, Row III (part 1)

S	kernel	coeff.	colour structure
S_7	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	$+N_c$	$U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$
S_7	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	$+1$	$f^{abc}\,f^{deh}\,U^{al}(w')\,U^{bd}(z)\,U^{ce}(w)\,\rho^l(x')\,\rho^h(x)$
S_7	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	$+1$	$f^{abc}\,f^{deh}\,U^{al}(w')\,U^{eb}(z)\,U^{cd}(x)\,\rho^l(x')\,\rho^h(x)$
S_7	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	-1	$f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(x)\,\rho^l(x')\,\rho^h(x)$
S_{10}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w',z)$	$+N_c$	$U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$
S_{10}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w',z)$	$+1$	$f^{abc}\,f^{deh}\,U^{al}(w)\,U^{bd}(z)\,U^{ce}(w')\,\rho^h(x')\,\rho^l(x)$
S_{10}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w',z)$	-1	$f^{abc}\,f^{deh}\,U^{ce}(x')\,U^{db}(z)\,U^{al}(w)\,\rho^l(x)\,\rho^h(x')$
S_{10}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w',z)$	$+1$	$f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(x')\,\rho^l(x)\,\rho^h(x')$
S_{12}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	$-N_c$	$U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$
S_{12}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	-1	$f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bl}(z)\,U^{cd}(w)\,\rho^h(x')\,\rho^l(x)$
S_{12}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	$+1$	$f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{lb}(z)\,U^{cd}(x')\,\rho^l(x)\,\rho^h(x')$

S	kernel	coeff.	colour structure
S_{12}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	-1	$f^{abc}f^{deh}\,U^{ad}(w)\,U^{ce}(x')\,U^{bl}(x)\,\rho^{l}(x)\,\rho^{h}(x')$
S_{13}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	$-N_c$	$U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')$
S_{13}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	$+1$	$f^{abc}f^{deh}\,U^{ad}(w)\,U^{bl}(z)\,U^{ce}(w')\,\rho^{l}(x)\,\rho^{h}(x')$
S_{13}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	-1	$f^{abc}f^{deh}\,U^{ce}(x')\,U^{lb}(z)\,U^{ad}(w)\,\rho^{h}(x')\,\rho^{l}(x)$
S_{13}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	$+1$	$f^{abc}f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(x)\,\rho^{h}(x')\,\rho^{l}(x)$

7.10 From Group III, Row V

S	kernel	coeff.	colour structure
S_{7}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	$-N_c$	$U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)$
S_{7}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	$+N_c$	$U^{ab}(w')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)$
S_{7}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	$+1$	$f^{abc}f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(x)\,\rho^{l}(x')\,\rho^{h}(x)$
S_{7}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	-1	$f^{abc}f^{deh}\,U^{ad}(w)\,U^{cl}(w')\,U^{be}(x)\,\rho^{l}(x')\,\rho^{h}(x)$
S_{8}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	$+N_c$	$U^{ab}(w')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)$
S_{8}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	$-N_c$	$U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)$
S_{8}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	-1	$f^{abc}f^{deh}\,U^{ad}(w)\,U^{bl}(x')\,U^{ce}(x)\,\rho^{l}(x')\,\rho^{h}(x)$
S_{8}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	$+1$	$f^{abc}f^{deh}\,U^{al}(w')\,U^{cd}(w)\,U^{be}(x)\,\rho^{l}(x')\,\rho^{h}(x)$
S_{9}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$	$+N_c$	$U^{ab}(x')\,U^{ac}(w)\,\rho^{b}(x')\,\rho^{c}(x)$
S_{9}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$	$-N_c$	$U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')$
S_{9}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$	$+1$	$f^{abc}f^{deh}\,U^{ae}(w')\,U^{bl}(x)\,U^{cd}(x')\,\rho^{l}(x)\,\rho^{h}(x')$
S_{9}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$	-1	$f^{abc}f^{deh}\,U^{al}(w)\,U^{ce}(w')\,U^{bd}(x')\,\rho^{h}(x')\,\rho^{l}(x)$
S_{10}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w',z)$	$-N_c$	$U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)$
S_{10}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w',z)$	$+N_c$	$U^{ab}(w)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)$
S_{10}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w',z)$	-1	$f^{abc}f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(x')\,\rho^{l}(x)\,\rho^{h}(x')$
S_{10}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w',z)$	$+1$	$f^{abc}f^{deh}\,U^{ae}(w')\,U^{cl}(w)\,U^{bd}(x')\,\rho^{h}(x')\,\rho^{l}(x)$
S_{12}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	$+N_c$	$U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)$
S_{12}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	$+1$	$f^{abc}f^{deh}\,U^{ae}(w')\,U^{bd}(x')\,U^{cl}(x)\,\rho^{h}(x')\,\rho^{l}(x)$
S_{12}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	$+1$	$f^{abc}f^{deh}\,U^{ad}(w)\,U^{ce}(x')\,U^{bl}(x)\,\rho^{l}(x)\,\rho^{h}(x')$
S_{12}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	-1	$f^{abc}f^{deh}\,U^{ad}(w)\,U^{ce}(w')\,U^{bl}(x)\,\rho^{l}(x)\,\rho^{h}(x')$
S_{13}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	$+N_c$	$U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')$
S_{13}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	-1	$f^{abc}f^{deh}\,U^{ad}(w)\,U^{be}(x')\,U^{cl}(x)\,\rho^{l}(x)\,\rho^{h}(x')$
S_{13}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	-1	$f^{abc}f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(x)\,\rho^{h}(x')\,\rho^{l}(x)$
S_{13}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	$+1$	$f^{abc}f^{deh}\,U^{ae}(w')\,U^{cd}(w)\,U^{bl}(x)\,\rho^{l}(x)\,\rho^{h}(x')$

7.11 From Group III, Row VI

S	kernel	coeff.	colour structure
S_{7}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	$-N_c$	$U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)$
S_{7}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	-1	$f^{abc}f^{deh}\,U^{al}(x')\,U^{bd}(z)\,U^{ce}(w)\,\rho^{l}(x')\,\rho^{h}(x)$
S_{7}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	-1	$f^{abc}f^{deh}\,U^{al}(x')\,U^{eb}(z)\,U^{cd}(x)\,\rho^{l}(x')\,\rho^{h}(x)$
S_{7}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	$+1$	$f^{abc}f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(x)\,\rho^{l}(x')\,\rho^{h}(x)$
S_{8}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	$-N_c$	$U^{ab}(w')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)$
S_{8}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	$+1$	$f^{abc}f^{deh}\,U^{da}(z)\,U^{eb}(x)\,U^{hl}(w')\,\rho^{l}(x')\,\rho^{c}(x)$
S_{8}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	$+N_c$	$U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)$
S_{8}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	$+1$	$f^{abc}f^{deh}\,U^{ae}(x)\,U^{db}(z)\,U^{cl}(x')\,\rho^{l}(x')\,\rho^{h}(x)$
S_{9}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$	$-N_c$	$U^{ab}(x')\,U^{ac}(w)\,\rho^{b}(x')\,\rho^{c}(x)$
S_{9}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$	$+1$	$f^{abc}f^{deh}\,U^{da}(z)\,U^{eb}(x')\,U^{hl}(w)\,\rho^{c}(x')\,\rho^{l}(x)$
S_{9}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$	$+N_c$	$U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')$
S_{9}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$	-1	$f^{abc}f^{deh}\,U^{ad}(x')\,U^{eb}(z)\,U^{cl}(x)\,\rho^{h}(x')\,\rho^{l}(x)$
S_{10}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w',z)$	$-N_c$	$U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)$
S_{10}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w',z)$	-1	$f^{abc}f^{deh}\,U^{al}(x)\,U^{bd}(z)\,U^{ce}(w')\,\rho^{h}(x')\,\rho^{l}(x)$
S_{10}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w',z)$	$+1$	$f^{abc}f^{deh}\,U^{al}(x)\,U^{db}(z)\,U^{ce}(x')\,\rho^{l}(x)\,\rho^{h}(x')$
S_{10}	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w',z)$	-1	$f^{abc}f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(x')\,\rho^{l}(x)\,\rho^{h}(x')$
S_{12}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	$+N_c$	$U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)$
S_{12}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	$+1$	$f^{abc}f^{deh}\,U^{ae}(x')\,U^{bl}(z)\,U^{cd}(w)\,\rho^{h}(x')\,\rho^{l}(x)$
S_{12}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	$-N_c$	$U^{ab}(z)\,U^{bc}(x')\,\rho^{a}(x)\,\rho^{c}(x')$
S_{12}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	$+1$	$f^{abc}f^{deh}\,U^{ad}(w)\,U^{ce}(x')\,U^{bl}(x)\,\rho^{l}(x)\,\rho^{h}(x')$
S_{13}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	$+N_c$	$U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')$
S_{13}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	-1	$f^{abc}f^{deh}\,U^{ad}(x')\,U^{bl}(z)\,U^{ce}(w')\,\rho^{l}(x)\,\rho^{h}(x')$
S_{13}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	$-N_c$	$U^{ab}(z)\,U^{bc}(x')\,\rho^{c}(x')\,\rho^{a}(x)$
S_{13}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	-1	$f^{abc}f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(x)\,\rho^{h}(x')\,\rho^{l}(x)$

7.12 From the four-ρ remainders d^3N_a, d^3N_b

S	kernel	coeff.	colour structure
S_1	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$	$-\frac{N_c}{2}$	$U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$
S_1	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$	$+\frac{N_c}{2}$	$U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$
S_1	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$	$-\frac{N_c}{2}$	$U^{ab}(w')\,U^{bc}(w)\,\rho^a(x')\,\rho^c(x)$
S_1	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$	$+\frac{N_c}{2}$	$U^{ab}(w)\,U^{ca}(x')\,\rho^c(x')\,\rho^b(x)$
S_2	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$	$-\frac{N_c}{2}$	$U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$
S_2	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$	$+\frac{N_c}{2}$	$U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$
S_2	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$	$+\frac{N_c}{2}$	$U^{ab}(w')\,U^{ca}(x)\,\rho^b(x')\,\rho^c(x)$
S_2	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$	$-\frac{N_c}{2}$	$U^{ab}(w')\,U^{cb}(w)\,\rho^a(x')\,\rho^c(x)$
S_3	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	-1	$f^{abc}\,f^{ade}\,U^{dh}(w')\,U^{hb}(w)\,\rho^e(x')\,\rho^c(x)$
S_3	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$+\frac{1}{2}$	$f^{abc}\,f^{ade}\,U^{hb}(w)\,U^{dh}(x')\,\rho^e(x')\,\rho^c(x)$
S_3	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$-\frac{1}{2}$	$f^{abc}\,f^{dae}\,U^{hb}(w')\,U^{dh}(x)\,\rho^c(x')\,\rho^e(x)$
S_3	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$-\frac{1}{2}$	$f^{abc}\,f^{deh}\,U^{eb}(w')\,U^{lc}(x)\,U^{ad}(x')\,\rho^h(x')\,\rho^l(x)$
S_3	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$+\frac{1}{2}$	$f^{abc}\,f^{deh}\,U^{lc}(x')\,U^{eb}(w)\,U^{ad}(x)\,\rho^l(x')\,\rho^h(x)$
S_{11}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$-\frac{1}{2}$	$f^{abc}\,f^{cde}\,U^{dh}(w')\,U^{hb}(w)\,\rho^a(x)\,\rho^e(x')$
S_{11}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$+\frac{1}{2}$	$f^{abc}\,f^{cde}\,U^{hb}(w)\,U^{dh}(x')\,\rho^a(x)\,\rho^e(x')$
S_{11}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$-\frac{1}{2}$	$f^{abc}\,f^{dce}\,U^{hb}(w')\,U^{dh}(x')\,\rho^a(x)\,\rho^e(x')$
S_{11}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$+\frac{1}{2}$	$f^{abc}\,f^{dce}\,U^{bh}(w')\,U^{dh}(w)\,\rho^a(x)\,\rho^e(x')$
S_{11}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$-\frac{1}{2}$	$f^{abc}\,f^{deh}\,U^{eb}(w')\,U^{dc}(x')\,U^{al}(x)\,\rho^l(x)\,\rho^h(x')$
S_{11}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$+\frac{1}{2}$	$f^{abc}\,f^{deh}\,U^{ec}(x')\,U^{db}(w)\,U^{al}(x)\,\rho^h(x')\,\rho^l(x)$
S_{11}	$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	$-N_c$	$U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$

8 The sum: JIMWLK

The two-ρ terms of Sec. 7 are added to two other sets: the genuine two-ρ terms of `Evolution_two_rho.tex` (with fixes F5 and F6), and the two-ρ lists of `4rho.tex` (with fixes F1 and F2). All of these are written out in `TwoRho_all.pdf`. Within each kernel structure, the colour structures are reduced to an independent basis, with exact coefficients that are polynomials in N_c.

- The column "3ρ list" is the total from Sec. 7.
- "genuine" and "`4rho.tex`" are the two other sets.
- The last column is JIMWLK⊗LO.

All 50 basis structures with a nonzero entry match. So, with the fixes and the operator order, the two-ρ terms coming from your three-ρ list complete the target evolution: the total coefficient of $\log(\vee/\Lambda)$ is JIMWLK⊗LO. Adding the $\log(\vee/k^+)$ terms of each row gives the coefficient of $\log\vee$, which is BFKL⊗LO (`TwoRho_all.pdf`, Sec. 9).

Structure S_1: $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$

colour structure	3ρ list	genuine	`4rho.tex`	sum	JIMWLK
$U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$	0	0	$2N_c$	$2N_c$	$2N_c$ ✓
$U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$	0	0	$-2N_c$	$-2N_c$	$-2N_c$ ✓
$f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(w')\,U^{hb}(x)\,\rho^l(x')\,\rho^c(x)$	1	0	1	2	2 ✓
$f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(x')\,U^{hb}(x)\,\rho^l(x')\,\rho^c(x)$	-1	0	-1	-2	-2 ✓

Structure S_2: $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$

colour structure	3ρ list	genuine	`4rho.tex`	sum	JIMWLK
$U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$	0	0	$2N_c$	$2N_c$	$2N_c$ ✓
$U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$	0	0	$-2N_c$	$-2N_c$	$-2N_c$ ✓
$f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(w)\,U^{hb}(x')\,\rho^c(x')\,\rho^l(x)$	1	0	1	2	2 ✓
$f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(x)\,U^{hb}(x')\,\rho^c(x')\,\rho^l(x)$	-1	0	-1	-2	-2 ✓

Structure S_3: $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$

colour structure	3ρ list	genuine	`4rho.tex`	sum	JIMWLK
$U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$	0	$\frac{N_c}{2}$	$\frac{3N_c}{2}$	$2N_c$	$2N_c$ ✓
$f^{abc}\,f^{ade}\,U^{hb}(x')\,U^{hd}(x)\,\rho^c(x')\,\rho^e(x)$	0	0	2	2	2 ✓

colour structure	3ρ list	genuine	`4rho.tex`	sum	JIMWLK
$f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(x)\,U^{hl}(x')\,\rho^l(x')\,\rho^c(x)$	-1	0	-1	-2	-2 ✓
$f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(x')\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)$	-1	0	-1	-2	-2 ✓
$f^{abc}\,f^{ade}\,U^{dh}(w')\,U^{hb}(w)\,\rho^e(x')\,\rho^c(x)$	-1	$\frac{1}{2}$	$\frac{1}{2}$	0	0 ✓
$f^{abc}\,f^{ade}\,U^{hb}(w)\,U^{dh}(x')\,\rho^e(x')\,\rho^c(x)$	1	0	-1	0	0 ✓
$f^{abc}\,f^{dae}\,U^{hb}(w')\,U^{dh}(x)\,\rho^c(x')\,\rho^e(x)$	-1	0	1	0	0 ✓
$f^{abc}\,f^{deh}\,U^{lc}(x')\,U^{eb}(w)\,U^{ad}(x)\,\rho^l(x')\,\rho^h(x)$	1	0	-1	0	0 ✓
$f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x')$	0	$\frac{1}{2}$	$-\frac{1}{2}$	0	0 ✓
$f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x)$	0	$-\frac{1}{2}$	$\frac{1}{2}$	0	0 ✓

Structure S_4: $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(w,w,z)$

colour structure	3ρ list	genuine	`4rho.tex`	sum	JIMWLK
$U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$	0	$-2N_c$	0	$-2N_c$	$-2N_c$ ✓
$U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$	0	$2N_c$	0	$2N_c$	$2N_c$ ✓
$f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(w')\,U^{hb}(w)\,\rho^l(x')\,\rho^c(x)$	0	-2	0	-2	-2 ✓
$f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(x')\,U^{hb}(w)\,\rho^l(x')\,\rho^c(x)$	0	2	0	2	2 ✓

Structure S_5: $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(w',w',z)$

colour structure	3ρ list	genuine	`4rho.tex`	sum	JIMWLK
$U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$	0	$-2N_c$	0	$-2N_c$	$-2N_c$ ✓
$U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$	0	$2N_c$	0	$2N_c$	$2N_c$ ✓
$f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(w)\,U^{hb}(w')\,\rho^c(x')\,\rho^l(x)$	0	-2	0	-2	-2 ✓
$f^{abc}\,f^{deh}\,U^{da}(z)\,U^{el}(x)\,U^{hb}(w')\,\rho^c(x')\,\rho^l(x)$	0	2	0	2	2 ✓

Structure S_6: $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(w,w',z)$

colour structure	3ρ list	genuine	`4rho.tex`	sum	JIMWLK
$U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$	0	$2N_c$	0	$2N_c$	$2N_c$ ✓
$f^{abc}\,f^{ade}\,U^{hb}(w')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)$	0	2	0	2	2 ✓
$f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(w)\,U^{hl}(w')\,\rho^l(x')\,\rho^c(x)$	0	-2	0	-2	-2 ✓
$f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(w')\,U^{hl}(w)\,\rho^c(x')\,\rho^l(x)$	0	-2	0	-2	-2 ✓

Structure S_7: $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$

colour structure	3ρ list	genuine	`4rho.tex`	sum	JIMWLK
$U^{ab}(x')\,U^{ca}(w)\,\rho^b(x')\,\rho^c(x)$	N_c	$-N_c$	0	0	0 ✓
$U^{ab}(w')\,U^{ca}(w)\,\rho^b(x')\,\rho^c(x)$	$-N_c$	N_c	0	0	0 ✓
$f^{abc}\,f^{deh}\,U^{al}(x')\,U^{bd}(z)\,U^{ce}(w)\,\rho^l(x')\,\rho^h(x)$	-1	1	0	0	0 ✓
$f^{abc}\,f^{deh}\,U^{al}(w')\,U^{bd}(z)\,U^{ce}(w)\,\rho^l(x')\,\rho^h(x)$	1	-1	0	0	0 ✓

Structure S_8: $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$

colour structure	3ρ list	genuine	`4rho.tex`	sum	JIMWLK
$U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$	$-N_c$	$-N_c$	0	$-2N_c$	$-2N_c$ ✓
$f^{abc}\,f^{ade}\,U^{hb}(w')\,U^{hd}(x)\,\rho^c(x')\,\rho^e(x)$	-2	0	0	-2	-2 ✓
$f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(x)\,U^{hl}(w')\,\rho^l(x')\,\rho^c(x)$	2	0	0	2	2 ✓
$f^{abc}\,f^{deh}\,U^{da}(z)\,U^{eb}(w')\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)$	1	1	0	2	2 ✓
$f^{abc}\,f^{dce}\,U^{hd}(w)\,U^{ah}(w')\,\rho^b(x')\,\rho^e(x)$	-1	1	0	0	0 ✓
$f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{be}(z)\,U^{cl}(w')\,\rho^l(x')\,\rho^h(x)$	-1	1	0	0	0 ✓

Structure S_9: $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$

colour structure	3ρ list	genuine	`4rho.tex`	sum	JIMWLK
$U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$	$-N_c$	$-N_c$	0	$-2N_c$	$-2N_c$ ✓
$f^{abc}\,f^{ade}\,U^{hb}(x')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)$	-2	0	0	-2	-2 ✓

colour structure	3ρ list	genuine	`4rho.tex`	sum	JIMWLK
$f^{abc}\, f^{deh}\, U^{da}(z)\, U^{eb}(w)\, U^{hl}(x')\, \rho^l(x')\, \rho^c(x)$	1	1	0	2	2 ✓
$f^{abc}\, f^{deh}\, U^{da}(z)\, U^{eb}(x')\, U^{hl}(w)\, \rho^c(x')\, \rho^l(x)$	2	0	0	2	2 ✓
$f^{abc}\, f^{cde}\, U^{hd}(w')\, U^{ah}(w)\, \rho^b(x)\, \rho^e(x')$	1	−1	0	0	0 ✓
$f^{abc}\, f^{deh}\, U^{ae}(w')\, U^{bd}(z)\, U^{cl}(w)\, \rho^l(x)\, \rho^h(x')$	1	−1	0	0	0 ✓

Structure S_{10}: $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\; K(x',w',z)$

colour structure	3ρ list	genuine	`4rho.tex`	sum	JIMWLK
$U^{ab}(x)\, U^{ca}(w')\, \rho^c(x')\, \rho^b(x)$	N_c	$-N_c$	0	0	0 ✓
$U^{ab}(w)\, U^{ca}(w')\, \rho^c(x')\, \rho^b(x)$	$-N_c$	N_c	0	0	0 ✓
$f^{abc}\, f^{deh}\, U^{al}(x)\, U^{bd}(z)\, U^{ce}(w')\, \rho^h(x')\, \rho^l(x)$	−1	1	0	0	0 ✓
$f^{abc}\, f^{deh}\, U^{al}(w)\, U^{bd}(z)\, U^{ce}(w')\, \rho^h(x')\, \rho^l(x)$	1	−1	0	0	0 ✓

Structure S_{11}: $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\; K(x,x',z)$

colour structure	3ρ list	genuine	`4rho.tex`	sum	JIMWLK

Structure S_{12}: $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\; K(x,w,z)$

colour structure	3ρ list	genuine	`4rho.tex`	sum	JIMWLK

Structure S_{13}: $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\; K(x,w',z)$

colour structure	3ρ list	genuine	`4rho.tex`	sum	JIMWLK

A Scripts

- `parse.py`: reads `input/three_rho_all.tex` (your list, verbatim) into terms.
- `fixes.py`: T1–T3, F3, F4. $\mathcal{D}$ in your notation is in `input/D_terms.tex`.
- `check3.py`, `cmp_entries.py`: row-by-row and term-by-term comparison with the earlier transcription.
- `check3b.py`: Table 1.
- `networks.py`: Sec. 5.
- `oporder.py`: the operator order of every term; check that it reproduces the two-ρ terms of the original expressions.
- `two_cmp.py`: Table 2.
- `make_note.py`: writes this file from `note_static.tex`.